

Republic of Tunisia
MINISTRY OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH



END OF STUDIES PROJECT REPORT

Presented to Obtain the National Diploma
in **Computer Engineering**

Subject:
Dynamic Email Composition Platform:
A Comprehensive Implementation with Vue.js and
Fastify for Effective Content Creation and
Management

Submitted By: Mohamed Khalil Bchir

Host Organisation: JoodLab

Professional Supervisor: Mr. Raed Bahri

Academic Supervisor: Mr. Yamen Mkaddem



Academic Year
2023-2024

Dedication

This work is dedicated to my dearest family, whose love and support have been the cornerstone of my journey.

To my parents, whose unwavering belief in me has been my greatest strength. Mom, your endless patience, and boundless love have been my guiding light. Dad, your wisdom and encouragement have been my anchor in the stormy seas of life. Your sacrifices and hard work have given me the foundation to reach for my dreams.

To my brothers, who have always been my first friends and constant companions. Your camaraderie, support, and belief in me have been a source of immense strength and joy. You are my pillars of strength, my partners in crime, and my best friends.

To my grandmother, whose warm hugs and wise words have been a source of comfort and inspiration. Your stories, your laughter, and your endless love have filled my life with immeasurable happiness. Your presence in my life is a blessing I cherish every day.

This dedication is a small token of my immense gratitude and love for you all. Your unwavering support has been my beacon of hope and inspiration. I could not have accomplished this without you. This is for you, with all my love.

Acknowledgments

I would like to express my deepest gratitude to those who have played a pivotal role in the completion of this project. Their support, guidance, and encouragement have been invaluable, and I am eternally grateful.

First and foremost, I extend my sincerest thanks to Mr.Yamen Mkaddem, my academic advisor. Your unwavering support, insightful feedback, and expert guidance have been instrumental in shaping this work. Your dedication to excellence and your passion for knowledge have inspired me throughout this journey.

I am profoundly grateful to Mr.Raed Bahri, my professional advisor. Your practical insights, profound expertise, and continuous encouragement have significantly enriched my learning experience. Your mentorship has been a cornerstone of my professional development, and your belief in my capabilities has been a source of great motivation.

A special mention goes to Madam Feyrouz Hamdaoui, whose experience and wisdom have been a guiding light. Your generous sharing of knowledge, thoughtful advice, and unwavering support have been truly invaluable. Your contributions have significantly enhanced the quality of this work, and for that, I am deeply thankful.

To all my colleagues and friends who have provided support and feedback, your contributions have been essential. Your camaraderie and encouragement have made this journey enjoyable and fulfilling.

Finally, I would like to thank my family for their unwavering love and support, without which none of this would have been possible.

To Mr.Yamen Mkaddem,Mr. Raed Bahri, and Madam Feyrouz Hamdaoui, your guidance and support will be remembered with deep gratitude for the rest of my life. Thank you for believing in me and for helping me achieve this milestone.

Contents

General Introduction	1
1 General presentation and study of the existing	3
Introduction	4
1.1 Presentation of the host company	4
1.2 Project context	4
1.2.1 Main issue	5
1.2.2 Analysis of the existing solutions	5
1.2.3 Overview of the Suggested Solution	7
1.3 Adopted Methodology	8
1.3.1 Methodology Choice	8
1.3.2 Introduction to SCRUM	8
1.3.3 SCRUM Roles	9
1.3.4 Product Backlog	9
1.3.5 The Sprint	10
Conclusion	11
2 Sprint 0: Requirements and specifications	12
Introduction	13
2.1 Email marketing tool Requirements	13
2.1.1 Identification of the actors	13
2.1.2 Functional Requirements	13
2.1.3 Non-functional requirements	14
2.2 Work Environment	15
2.2.1 Software Environment	15
2.2.2 Development Environment	18
2.3 Project Management with Scrum	21
2.3.1 Backlog Features	21

2.3.2	Global Use Case Diagram	22
2.3.3	Sprint Planning	23
2.4	General Architecture of the Application	24
2.4.1	Physical Architecture	24
2.4.2	Logical Architecture	25
	Conclusion	26
3	Study and Implementation of Sprint 1	27
	Introduction	28
3.1	Sprint 1 Backlog	28
3.2	Functional Specification	29
3.2.1	Use Case Diagram	29
3.2.2	Textual Description of Use Cases	30
3.3	Design	32
3.3.1	Class Diagram	32
3.3.2	Sequence Diagrams	33
3.4	Implementation	38
3.4.1	Authentication Interface	38
3.4.2	Email Template Builder	38
3.4.3	Email Template Library	39
3.4.4	Campaigns Publishing and Creation	39
3.4.5	Campaigns List	40
	Conclusion	41
4	Study and Implementation of Sprint 2	42
	Introduction	43
4.1	Sprint 2 Backlog	43
4.2	Functional Specification	44
4.2.1	Use Case Diagram	44
4.2.2	Textual Description of Use Cases	45
4.3	Design	47
4.3.1	Class Diagram	47
4.3.2	Sequence Diagrams	48

4.4	Implementation	54
4.4.1	Upload Media	54
4.4.2	Create New Organization	54
4.4.3	Invite New Members to Join Organization	55
4.4.4	Manage Members And Roles Within an Organization	55
	Conclusion	56
5	Study and Implementation of Sprint 3	57
	Introduction	58
5.1	Sprint 3 Backlog	58
5.2	Functional Specification	58
5.2.1	Use Case Diagram	59
5.2.2	Textual Description of Use Cases	59
5.3	Design	60
5.3.1	Class Diagram	60
5.3.2	Sequence Diagrams	61
5.4	Deployment	63
5.4.1	Deployment Process	63
5.4.2	Deployment Diagram	64
5.4.3	CI/CD Pipeline Diagram	64
5.5	Implementation	64
5.5.1	Insights Dashboard	65
	Conclusion	66
	General Conclusion	67
	References	68

List of Figures

1.1	Joodlab logo	4
1.2	HubSpot logo	5
1.3	Constant Contact logo	6
1.4	Life-cycle of the Scrum method	8
2.1	Visual Studio Code logo	15
2.2	GitLab logo	16
2.3	Notion logo	16
2.4	Postman logo	16
2.5	DBeaver logo	17
2.6	Docker Desktop logo	17
2.7	Git logo	17
2.8	Brevo logo	18
2.9	Node.js logo	18
2.10	TypeScript logo	18
2.11	Vue 3 logo	18
2.12	Tailwind CSS logo	19
2.13	Fastify logo	19
2.14	MJML logo	19
2.15	Turborepo logo	20
2.16	Prisma logo	20
2.17	Swagger logo	20
2.18	Global Use Case Diagram	22
2.19	Sprint Planning	23
2.20	Global Use Case Diagram	24
2.21	Global Use Case Diagram	25
3.1	Sprint 1 Use Case Diagram	29

3.2	Sprint 1 Class Diagram	32
3.3	Log In Sequence Diagram	33
3.4	Register Sequence Diagram	34
3.5	Create Email Template Sequence Diagram	35
3.6	Create And Publishing Campaigns Sequence Diagram	36
3.7	Create Audience Sequence Diagram	37
3.8	User Interface for Authentication Interface	38
3.9	User Interface for Email Template Builder	39
3.10	Email Template Library	39
3.11	User Interface for Campaigns Publishing and Creation	40
3.12	User Interface for Campaigns List	40
4.1	Sprint 2 Use Case Diagram	44
4.2	Sprint 2 Class Diagram	47
4.3	Media Upload Sequence Diagram	48
4.4	Organization Creation Sequence Diagram	49
4.5	Send Invitation Sequence Diagram	50
4.6	Accept Invitation Sequence Diagram	51
4.7	Assign Roles Sequence Diagram	52
4.8	Remove Members Sequence Diagram	53
4.9	User Interface for Upload Media	54
4.10	User Interface for Creating a New Organization	55
4.11	User Interface for Inviting New Members	55
4.12	User Interface for Assigning Roles And Managing Members	56
5.1	Sprint 3 Use Case Diagram	59
5.2	Sprint 3 Class Diagram	60
5.3	Tracking Sequence Diagram	61
5.4	Tracking Sequence Diagram	62
5.5	Deployment Diagram	64
5.6	CI/CD Pipeline Diagram	64
5.7	User Interface for Campaigns Dashboard	65

List of Tables

1.1	Comparison of HubSpot, Constant Contact, and our Email Marketing Tool	8
3.1	Sprint 1 Backlog	28
3.2	Use Case 1: Register	30
3.3	Use Case 2: Authenticate	30
3.4	Use Case 3: Create Email Template	31
3.5	Use Case 4: Create Campaign	31
3.6	Use Case 5: Create Audience	31
4.1	Sprint 2 Backlog	43
4.2	Use Case 1: Upload Media Files	45
4.3	Use Case 2: Create Organizations	45
4.4	Use Case 3: Assign Roles to Users	46
4.5	Use Case 4: Send Invitation to Join Organization via Email	46
4.6	Use Case 5: Remove User from an Organization	47
5.1	Sprint 3 Backlog	58
5.2	Use Case 1: View Insights	60

Liste of acronymes

SaaS Software as a service

MJML Mailjet Markup Language

JWT Json Web Token

General Introduction

In the digital age, email marketing has emerged as a crucial tool for businesses. It provides a platform for businesses to reach out to their customers directly, offering personalized content and promotions that can drive both engagement and sales. This direct line of communication, when coupled with the right messaging, can transform email into one of the most impactful marketing channels. It's a chance for businesses to speak directly to their customers in their inbox, at a time that is convenient for them.

Consider the case of Amazon, a global e-commerce giant. Amazon has effectively harnessed the power of email marketing, sending personalized product recommendations and offers to its customers. This strategy significantly enhances customer engagement and drives sales, demonstrating the potential of email marketing. It's a testament to how understanding customer preferences and tailoring messages accordingly can lead to increased customer loyalty and revenue.

However, like any other marketing strategy, email marketing comes with its own set of challenges. These include creating engaging content that resonates with the audience, ensuring email deliverability despite the presence of rigorous spam filters, maintaining email lists to ensure they are up-to-date and relevant, and complying with regulations like the General Data Protection Regulation (GDPR). These challenges can make email marketing seem daunting, but they are not invincible.

To overcome these challenges, businesses are turning to solutions like email builders. These tools provide a seamless and intuitive platform for creating compelling email campaigns. They come equipped with features like drag-and-drop editors, pre-designed templates, and automation tools that make the process of crafting and sending emails a breeze. These tools not only enhance user engagement but also increase brand visibility.

Moreover, these solutions often come with built-in analytics tools. These tools allow businesses to track key metrics like open rates, click-through rates, and conversion rates. By analyzing these metrics, businesses can gain insights into what's working and what's

not in their email campaigns, allowing them to continuously optimize their strategies for better results.

In conclusion, email marketing is a powerful tool for businesses in the digital age. Despite challenges, with the right strategies and tools, it can drive growth and success. Businesses that effectively use email marketing are more likely to succeed. Its influence in the marketing world is significant and will continue to be a dominant force for years to come.

This report will delve into various aspects of the email marketing tool developed during the project. The structure of the report is as follows:

- **Chapter 1: General Presentation and Study of Existing Solutions**

This chapter provides an overview of email marketing and examines existing tools and solutions available in the market.

- **Chapter 2: Requirements and Specifications**

This chapter outlines the requirements and specifications for the development of the email marketing tool, detailing the necessary features and functionalities.

- **Chapter 3: Email Builder Feature and Campaign Publishing**

This chapter focuses on the email builder feature, explaining how users can create email templates and publish campaigns using the tool.

- **Chapter 4: Workspace Management and Media Cloud Feature**

This chapter discusses the workspace management feature, which allows for the creation and organization of different workspaces. It also covers the media cloud feature for managing media files.

- **Chapter 5: Campaign Tracking Feature**

The final chapter describes the campaign tracking feature, which enables users to monitor the performance of their email campaigns through various metrics and analytics.

The subsequent chapters will provide a detailed exploration of each of these aspects, demonstrating the capabilities and functionalities of the developed email marketing tool.

1

General presentation and study of the
existing

Introduction

In this chapter, we will first introduce the host company, Joodlab. Then, we will give an overview of the general context of our project and explain the working methodology that we will adopt in our project.

1.1 Presentation of the host company

Joodlab is a technology company that provides the simplest solutions and technical consultations for various companies in different sectors. It was established in Saudi Arabia by a group of experts with multiple competencies in the fields of information technology, artificial intelligence, and information automation.



Figure 1.1: Joodlab logo

1.2 Project context

In this part, we will present the main issue, a study of the existing solutions, our project's goals as well as the different steps of the project.

1.2.1 Main issue

The primary challenge in the field of email marketing is the creation and management of effective email campaigns. This involves several aspects:

- **Email Creation:** The process of creating an email should be intuitive and efficient. Currently, many tools require knowledge of HTML and CSS for customization, which can be a barrier for non-technical users.
- **Customization:** Users should be able to easily customize their emails to fit their brand and message. This includes the ability to change styles, add elements, and adjust layouts.
- **Campaign Management:** Managing an email campaign involves sending out emails, scheduling them, and segmenting the audience. These tasks should be easy to perform and manage within the tool.
- **Tracking:** Once an email campaign is sent out, it's crucial to track its performance. This includes open rates, click-through rates, and conversions. The tool should provide comprehensive tracking and analytics.
- **Usability:** The tool should be user-friendly, with a clean interface and clear instructions. It should cater to both beginners and advanced users, offering simplicity without compromising on features.

The lack of a tool that effectively addresses all these aspects is the main issue we aim to solve with our SaaS email marketing application.

1.2.2 Analysis of the existing solutions

HubSpot



Figure 1.2: HubSpot logo

While HubSpot is a comprehensive platform for email marketing, it has several limitations that can affect its usability and effectiveness:

- **Limited Free Version:** The free version of HubSpot’s email marketing tool is limited. You can only send 2,000 emails per month. Additionally, you cannot connect to a custom email sending domain. This means your emails will show “via HubSpot” in the sender area and the footer of your email will have a HubSpot logo.
- **Sending Limits:** HubSpot’s send limits vary based on each email provider’s limit. For instance, Gmail Free allows 350 emails per day, Google Apps, Office 365, Exchange inbox, and Generic inbox allow 1000 emails per day.
- **CRM Limitations:** There are major limitations for sales functionality in the free CRM. These include only 15 minutes of calling per user, per month, no inbound calling, HubSpot branding on emails, only one sales pipeline, and no email automation.

Constant Contact



Figure 1.3: Constant Contact logo

Constant Contact, despite its popularity, also has its drawbacks:

- **Inflexible Email Editor:** The email editor is not as flexible or intuitive as users might prefer, which can make the process of creating emails more time-consuming and less efficient.
- **Limited Tracking Features:** The tracking features, while useful, do not offer as much depth as other tools. This can limit user’s ability to fully understand and optimize their email campaigns.

In summary, both HubSpot and Constant Contact have significant limitations that can hinder the efficiency and effectiveness of email marketing. Our proposed solution aims to address these limitations by offering a more intuitive and flexible email builder, more customizable design options, and more accessible yet comprehensive campaign tracking.

1.2.3 Overview of the Suggested Solution

Our proposed solution is a Software as a Service (SaaS) application designed to revolutionize email marketing. The key features of our solution include:

- **Drag-and-Drop Email Builder:** Our solution includes a user-friendly drag-and-drop email builder. This allows users to easily create professional and personalized emails without needing any technical skills. The builder includes a variety of elements that can be added to an email, such as text boxes, images, buttons, and more.
- **Customizable Elements:** Each element in the email builder is fully customizable. Users can change the style, size, color, and other properties of these elements to fit their brand and message. This level of customization allows for the creation of unique and engaging emails.
- **Email Sending and Campaign Creation:** Our solution makes it easy to send out emails and create marketing campaigns. Users can schedule emails to be sent at specific times, segment their audience to target specific groups, and automate their email marketing process.
- **Campaign Tracking:** To help users measure the success of their email marketing efforts, our solution includes comprehensive campaign tracking features. Users can track open rates, click-through rates, and conversions, and use this data to optimize their future campaigns.
- **User-Friendly Interface:** Our solution features a clean and intuitive interface, making it easy for users to navigate and use the various features. This ensures a smooth user experience, regardless of the user's technical skills.

In summary, our SaaS application aims to provide a more intuitive, efficient, and effective approach to email marketing. By addressing the limitations of existing solutions,

Feature	HubSpot	Constant Contact	Our Tool
Tracking	Yes	Yes	Yes
Campaign Management	Yes	Yes	Yes
Email Builder	Limited	Limited	Yes
Customization	Limited	Limited	Yes

Table 1.1: Comparison of HubSpot, Constant Contact, and our Email Marketing Tool

we believe our tool can significantly improve the email marketing process for businesses of all sizes.

1.3 Adopted Methodology

In this section, we will discuss the methodology adopted for the development of our SaaS email marketing application.

1.3.1 Methodology Choice

For this project, we have chosen to adopt the SCRUM methodology [1]. This decision was made due to several reasons. Firstly, SCRUM is an agile development methodology which allows for rapid iteration and flexibility. This is particularly important for our project as it allows us to quickly adapt to changes and continuously improve our product based on user feedback. Secondly, SCRUM promotes close collaboration among team members which helps to ensure that everyone is on the same page and working towards the same goals.

1.3.2 Introduction to SCRUM

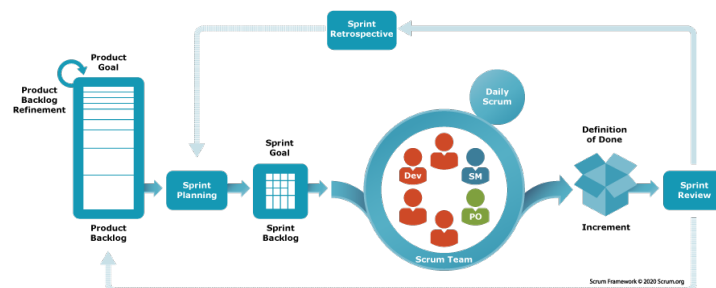


Figure 1.4: Life-cycle of the Scrum method

SCRUM is a framework for agile development. It is designed to add energy, focus, clarity, and transparency to project planning and implementation. At its core, SCRUM is about empowering a self-managing team to deliver, and it uses iterative, incremental practices to increase productivity and optimize predictability. In the context of our project, SCRUM provides a structured approach to managing the complex task of developing a new software product [1].

1.3.3 SCRUM Roles

Within the SCRUM framework, there are three key roles: the Product Owner, the Scrum Master, and the Development Team.

- **Product Owner:** The Product Owner is responsible for maximizing the value of the product resulting from the work of the Development Team. They manage the Product Backlog and are the sole person responsible for its content, availability, and ordering [1].
- **Scrum Master:** The Scrum Master is responsible for promoting and supporting SCRUM. They do this by helping everyone understand SCRUM theory, practices, rules, and values. The Scrum Master serves the team by removing impediments to the team's progress [1].
- **Development Team:** The Development Team is responsible for delivering potentially shippable increments of the product at the end of each Sprint. The team is self-organizing, cross-functional, and has all the skills necessary to create a product Increment [1].

1.3.4 Product Backlog

The Product Backlog is a list of features, functions, requirements, enhancements, and fixes that constitute the changes to be made to the product in future releases. It is the single source of requirements for any changes to be made to the product. The Product Owner is responsible for the Product Backlog, including its content, availability, and ordering [1].

- **Content:** The Product Backlog lists all features, functions, requirements, enhancements, and fixes that need to be made to the product. Each item in the backlog is expressed as a user story which provides a clear understanding of the requirement from an end-user perspective [1].
- **Availability:** The Product Backlog is a living document that is available to all stakeholders. This ensures transparency and allows everyone involved in the project to understand the direction of the product and the work to be done [1].
- **Ordering:** The items in the Product Backlog are ordered by the Product Owner based on their value, risk, priority, and necessity. This helps the Development Team to understand the work items' relative priority [1].

1.3.5 The Sprint

A Sprint is a time-boxed period during which specific work has to be completed and made ready for review. Sprints typically last between one week to one month. Here are the key aspects of a Sprint [1]:

- **Sprint Planning:** At the start of the Sprint, the team holds a planning meeting to decide what work will be accomplished during the Sprint. The Product Owner identifies the items in the Product Backlog that he or she wants completed (the Sprint Goal) and the team decides how much of this they can commit to complete during the next Sprint [1].
- **Daily Scrum:** Each day during the Sprint, the team holds a daily scrum meeting to discuss progress and plan for the day. This is not a problem-solving or issue resolution meeting. Instead, it is a chance for the team to inspect their progress towards the Sprint Goal and adjust their plan as necessary [1].
- **Sprint Review:** At the end of the Sprint, the team holds a Sprint Review to inspect the Increment and adapt the Product Backlog if needed. During the Sprint Review, the Scrum Team and stakeholders collaborate about what was done in the Sprint [1].
- **Sprint Retrospective:** After the Sprint Review and prior to the next Sprint Planning, the team holds a Sprint Retrospective. During this meeting, the team discusses

what went well, what didn't, and how they can make improvements for the next Sprint [1].

Conclusion

In conclusion, our SaaS email marketing tool, with its user-friendly drag-and-drop email builder and customizable elements, aims to address the limitations of existing solutions such as HubSpot. By adopting the SCRUM methodology, we ensure a flexible and iterative development process, allowing for continuous product improvement based on user feedback.

The SCRUM roles of Product Owner, Scrum Master, and Development Team will work collaboratively throughout the development process. The use of a Product Backlog and Sprints will effectively manage tasks and progress.

Overall, we believe our approach will lead to the successful development of a tool that significantly improves the email marketing process, providing value to our users and setting a new industry standard.

2

Sprint 0: Requirements and specifications

Introduction

This chapter starts with a brief introduction to the project. We'll then identify the key players and requirements. Next, we'll model the functional requirements and plan the product backlog and sprints. Finally, we'll discuss the development environment for the project.

2.1 Email marketing tool Requirements

This section outlines the requirements for an email marketing tool. The tool is designed to meet the needs of various actors. The requirements are divided into functional and non-functional requirements, providing a comprehensive overview of what the system is expected to do and the standards it should adhere to.

2.1.1 Identification of the actors

The system will interact with the following actors:

- **New User:** A person who has not registered an account yet.
- **User:** A person who has registered an account and can log in.
- **Administrator:** A user who has created an organization and manages it.

2.1.2 Functional Requirements

New Users

- **Account Registration**

Users

- **Account Login**
- **Organization Creation**
- **Organization List View**
- **Email Template Creation**

- **Email Template Editing**
- **Email Template List View**
- **Email Template Search**
- **Draft Management**
- **Email Sending**
- **Sent Email History**
- **Campaign Creation**
- **Campaign Management**
- **Campaign List View**
- **Campaign Insights**
- **Media Cloud**

Administrators

- **User Deactivation/Removal**
- **User Addition and Role Assignment**
- **Role Management**

2.1.3 Non-functional requirements

The non-functional requirements of the system are as follows:

- **Security:** The system should securely store user credentials and personal information. It should also provide secure access control mechanisms.
- **Usability:** The system should be user-friendly, allowing users to easily navigate and manage their organizations and email templates.
- **Performance:** The system should provide fast and efficient access to the shared work within an organization.

- **Reliability:** The system should reliably deliver emails and track their status and timestamp.
- **Scalability:** The system should be able to handle a large number of users, organizations, and email templates without performance degradation.
- **Maintainability:** The system should be easy to update and maintain, allowing for the addition of new features and updates to existing ones.

2.2 Work Environment

This section outlines the work environment for the project, including the hardware environment, development environment, and software environment.

2.2.1 Software Environment

In our project, we used the following tools:

Visual Studio Code



Figure 2.1: Visual Studio Code logo

The development environment for this project is a combination of the software tools mentioned in the Software Environment section. Each tool plays a crucial role in different aspects of the development process [2].

GitLab



Figure 2.2: GitLab logo

GitLab serves as the remote repository for code hosting and versioning. It also provides a platform for creating CI/CD pipelines, facilitating continuous integration and deployment of the project [3].

Notion



Figure 2.3: Notion logo

Notion serves as the central hub for project management. It is used for task tracking, documentation, and collaboration, providing a unified workspace for all project-related information and tasks [4].

Postman



Figure 2.4: Postman logo

Postman is used for testing and debugging APIs. It simplifies the process of building, testing, and modifying APIs, making it an essential tool for backend development [5].

DBeaver



Figure 2.5: DBeaver logo

DBeaver is used for database management and interaction. It supports a wide range of databases and provides a user-friendly interface for managing database schemas, running SQL queries, and performing other database-related tasks [6].

Docker Desktop

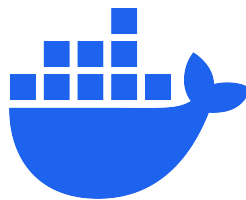


Figure 2.6: Docker Desktop logo

Docker Desktop is used to containerize the services. It allows for the packaging and distribution of applications in a manner that is independent of the host operating system, ensuring consistent operation across different computing environments [7].

Git



Figure 2.7: Git logo

Git is used for version control, allowing for efficient tracking and management of changes to the project codebase. It is an essential tool for collaborative development [8].

Brevo



Figure 2.8: Brevo logo

Brevo was utilized for transactional webhook and email services, providing reliable communication channels for real-time updates and notifications within the application [9].

2.2.2 Development Environment

Node.js



Figure 2.9: Node.js logo

Node.js is a JavaScript runtime built on Chrome's V8 JavaScript engine. It's used for building scalable network applications and executing JavaScript code server-side [10].

TypeScript



Figure 2.10: TypeScript logo

TypeScript is a typed superset of JavaScript that adds static types. It helps to write more robust code and maintain a cleaner and more consistent codebase. It's used in this project along with Vue 3 for building user interfaces [11].

Vue 3



Figure 2.11: Vue 3 logo

Vue 3 is the latest version of Vue.js, a progressive JavaScript framework for building user interfaces [12].

Tailwind CSS



Figure 2.12: Tailwind CSS logo

Tailwind CSS is a utility-first CSS framework for rapidly building custom user interfaces. It provides low-level utility classes that let you build completely custom designs without ever leaving your HTML [13].

Fastify



Figure 2.13: Fastify logo

Fastify is a web framework highly focused on providing the best developer experience with the least overhead and a powerful plugin architecture. It is used for building efficient and scalable Node.js web applications [14].

MJML



Figure 2.14: MJML logo

MJML is a markup language designed to simplify the process of coding responsive emails. It provides a semantic syntax and a library of standard components, making it easier to build custom email templates that are responsive and compatible with various email clients [15].

Turborepo



Figure 2.15: Turborepo logo

Turborepo is a high-performance build system for JavaScript and TypeScript codebases. It provides features like incremental builds, parallel execution, and remote caching. It's designed for monorepos, improving build times and facilitating efficient project management [16].

Prisma



Figure 2.16: Prisma logo

Prisma is an open-source database toolkit. It replaces traditional ORMs and makes database access easy with an auto-generated and type-safe query builder that's tailored to your database schema [17].

Swagger



Figure 2.17: Swagger logo

Swagger is an open-source software framework backed by a large ecosystem of tools that helps developers design, build, document, and consume RESTful web services. It's used in this project for API design and documentation [18].

2.3 Project Management with Scrum

This section outlines the key aspects of project management with Scrum including the product backlog, use case diagram, sprint planning, and interface prototyping.

2.3.1 Backlog Features

ID	Description	Effort
1	Register for an account with a password and email to access the system.	Low
2	Log in to the account using email and password for secure access.	Low
3	Reset the password via email in case of forgetting it.	Low
4	Create a new organization with a unique name and become its administrator to manage users and templates.	High
5	View a list of all organizations to navigate and manage them.	Medium
6	Access shared work (templates, campaigns, media) within the organization for effective collaboration.	High
7	Deactivate or remove users from the organization for security and access control.	Medium
8	Add users to the organization and assign specific roles for access control.	High
9	View and manage user roles within the organization.	Medium
10	Create new email templates for sending emails.	High
11	Edit and update email templates to keep information accurate.	Medium
12	View a list of all email templates for easy management.	Low
13	Create and manage email template drafts before finalizing and sending.	High
14	Create and publish campaigns using selected templates to organize and track marketing efforts..	High
15	View a history of published campaigns for reference.	Low
16	Generate reports for campaign analysis and informed decision-making.	High

2.3.2 Global Use Case Diagram

Figure 2.21 offer us a global overview by presenting a visual description of the functional behaviour of our tool. This diagram sums up the interactions among the actors and the diverse use cases within the system.

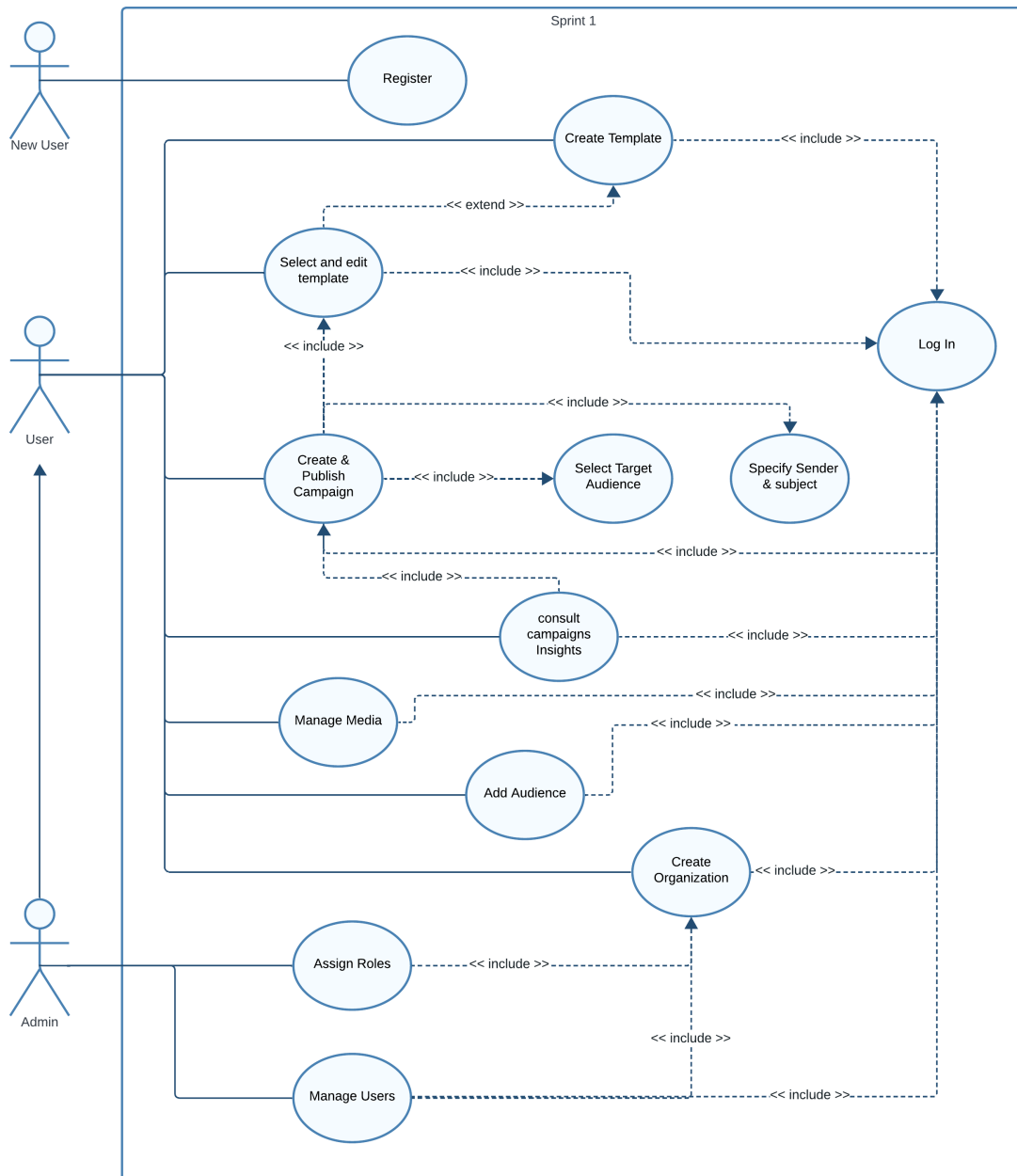


Figure 2.18: Global Use Case Diagram

2.3.3 Sprint Planning

Figure 2.19 outlines our project’s sprint planning. Each sprint had specific goals and durations. The initial phase was for learning the necessary tools. **Sprint 1** focused on building the email builder and creating and managing campaigns and audience. **Sprint 2** worked on media and organization management. **Sprint 3** focused on the deployment process and tracking functionality. This structured approach ensured thorough development and implementation of each feature. As a result, all planned features were successfully implemented within the project timeline.



Figure 2.19: Sprint Planning

2.4 General Architecture of the Application

This section outlines the general architecture of the application. It includes the physical architecture and the operation of the architecture.

2.4.1 Physical Architecture

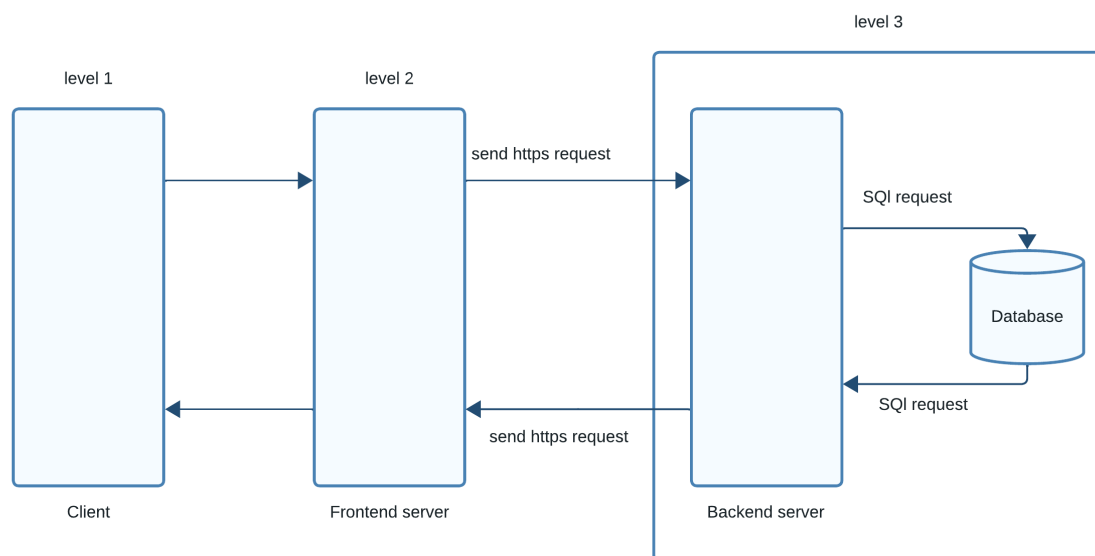


Figure 2.20: Global Use Case Diagram

In the physical architecture, I followed the 3-tier architecture:

- Level 1 is the Client (Presentation Tier): This is where the user interacts with the system. It could be any user interface like a web page, desktop application, or mobile app.
- Level 2 is the Front Server (Application Tier): This is where all the business logic resides. It processes client requests, manages business rules, and controls application functionality.
- Level 3 is the Back Server and Database (Data Tier): This is where data is stored and retrieved. It includes the database management system and the servers that manage it.

2.4.2 Logical Architecture

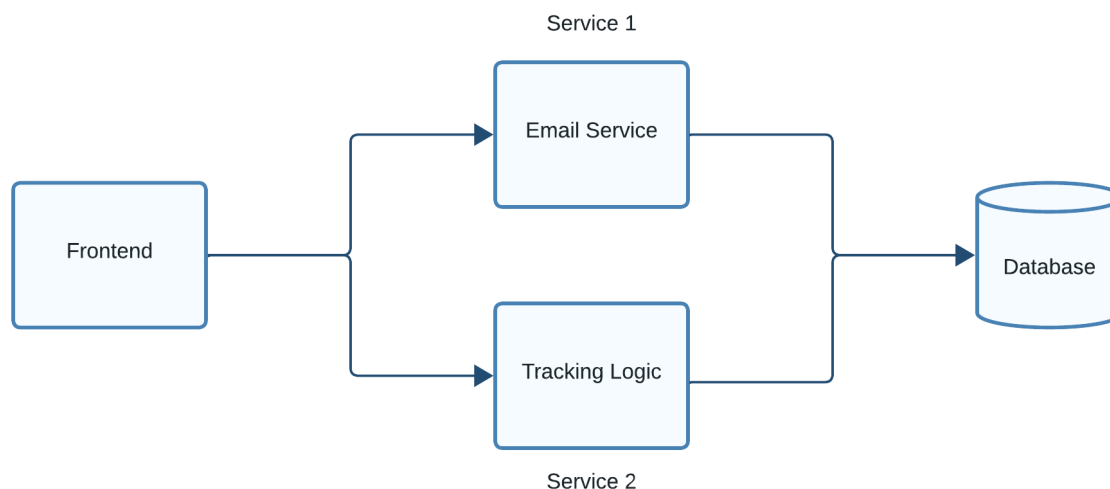


Figure 2.21: Global Use Case Diagram

The logical architecture of our email marketing tool consists of several interconnected components:

- **Frontend:**
 - This component represents the user-facing part of our application.
 - It is deployed on Vercel, a platform for hosting frontend applications.
 - Users interact with the frontend to create and manage email campaigns, templates, and other settings.
- **Email Building and Sending Service:**
 - Responsible for email composition, rendering, and sending logic.
 - Communicates with the frontend to receive email content and recipient lists.
 - Handles the process of sending emails to recipients.
 - It is deployed on Azure.
- **Tracking Service:**
 - Tracks email campaign performance metrics.
 - Collects data on email opens, clicks, and other relevant events.

- Stores tracking data for analytics and reporting purposes.
- It is deployed on Azure.
- **Database:**
 - Hosts our Campaign-related data.
 - Contains three main tables:
 - **Email Campaigns Data:** Stores email content, and recipient information.
 - **Email Template:** Stores built email templates
 - **Tracking Data:** Stores tracking events (e.g., opens, clicks) associated with email campaigns.

Conclusion

In this chapter, we identified the functional and non-functional requirements of our application. Then, we aligned scrum to our project. Finally, we chose our working environment.

3

Study and Implementation of Sprint 1

Introduction

This chapter provides an overview of the first sprint in our project. We start by defining the tasks in our Sprint 1 Backlog. Next, we present a detailed Functional Specification, which includes a Use Case Diagram and a Textual Description of Use Cases. Moving on to the Design aspect, we provide Detailed Sequence Diagrams and a Class Diagram. In the Implementation section, we discuss the creation of an Authentication Interface and an Interface for the Email Builder and an Interface for Creating and Publishing Campaigns

3.1 Sprint 1 Backlog

The table below outlines the tasks planned for Sprint 1. This sprint focuses on the development of key features for the email marketing tool, including the email builder, email sending functionalities, and user registration and authentication.

ID	Task Description
1	Create a user registration system, including the interface and backend functionality for new users to sign up.
2	Implement user authentication, enabling registered users to log in securely.
3	Enable access to shared resources (templates, campaigns, media) within an organization, promoting effective collaboration.
4	Implement the email builder feature. This involves creating the user interface and functionality for customizing email templates.
5	Create customizable email templates for various communication needs.
6	Enable users to edit and update email templates to keep information accurate and relevant.
7	Allow users to create and manage email template drafts before finalizing and sending.
8	Develop the email sending logic, allowing users to send emails using selected templates to specified recipients.
9	Implement campaign creation and publishing using selected templates.
10	Maintain a history of published campaigns for reference and analysis.

Table 3.1: Sprint 1 Backlog

The structured approach outlined in this backlog ensures clarity in understanding the responsibilities and the goals for this sprint.

3.2 Functional Specification

This section provides a comprehensive overview of the system's functionality. It includes a use case diagram illustrating the system's interactions and a detailed textual description of each use case.

3.2.1 Use Case Diagram

Figure 3.1 offers us an overview by presenting a visual description of the functional behaviour of our tool during the first sprint. This diagram sums up the interactions among the actors and the diverse use cases within the system.

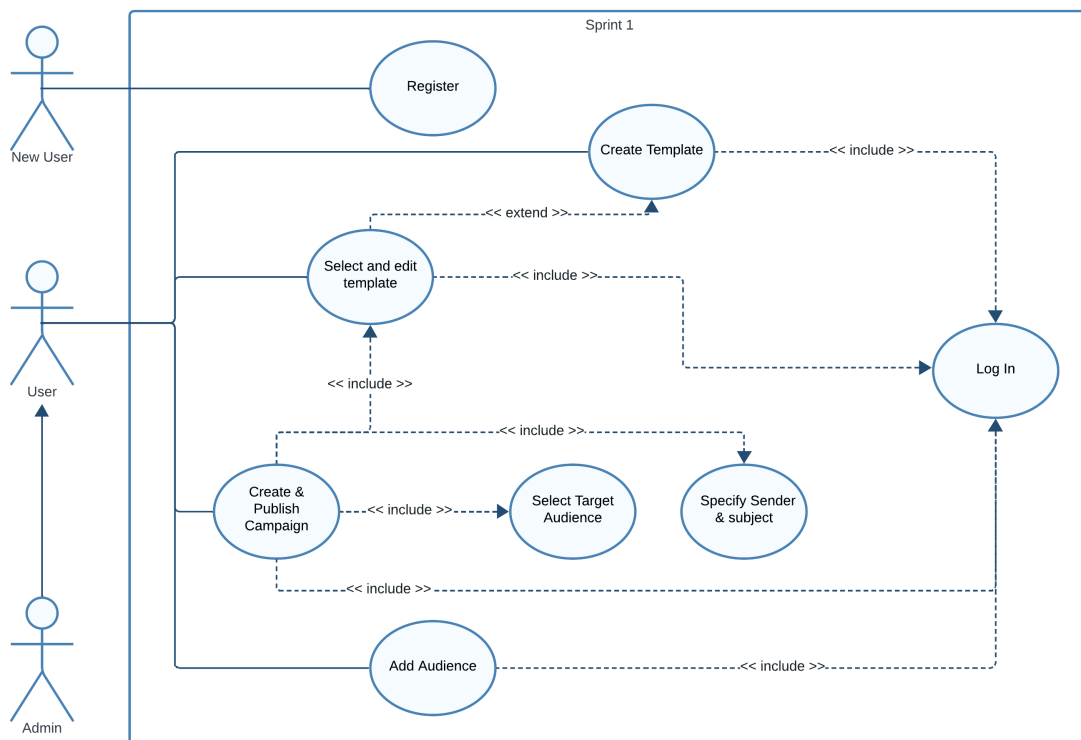


Figure 3.1: Sprint 1 Use Case Diagram

3.2.2 Textual Description of Use Cases

The following tables provide a detailed description of the use cases. This includes the actor, the goal of the use case, the preconditions, the basic flow, the alternative flows, and the postconditions.

Use Case 1: Register

Table 3.2 presents a textual description of a use case: Register.

Actor	New User
Goal	To register a new user to the system
Preconditions	The new user has accessed the registration page
Basic Flow	1. The new user enters their details in the registration form 2. The system validates the entered details 3. The system creates a new user account
Alternative Flows	If the entered details are invalid, the system shows an error message
Postconditions	A new user account is created and the new user can log in

Table 3.2: Use Case 1: Register

Use Case 2: Log In

Table 3.3 presents a textual description of a use case: Log In.

Actor	User, Admin
Goal	To authenticate a user to the system
Preconditions	The user has a registered account and has accessed the login page
Basic Flow	1. The user enters their login credentials 2. The system validates the entered credentials 3. The system logs the user in
Alternative Flows	If the entered credentials are invalid, the system shows an error message
Postconditions	The user is authenticated and logged in to the system

Table 3.3: Use Case 2: Authenticate

Use Case 3: Create Email Template

Table 3.4 presents a textual description of a use case: Create Email Template.

Actor	User, Admin
Goal	To create a new email template
Preconditions	The user is authenticated and has accessed the email builder
Basic Flow	1. The user selects the option to create a new template 2. The user designs the template and saves it
Alternative Flows	None
Postconditions	A new email template is created and saved in the system

Table 3.4: Use Case 3: Create Email Template

Use Case 4: Create Campaign

Table 3.5 presents a textual description of a use case: Create Campaign.

Actor	User, Admin
Goal	To Create and publish a campaign
Preconditions	The user is authenticated, has selected an email template, and has specified the recipient, the sender
Basic Flow	1. The user clicks the "save and publish" button 2. The system sends the email to the specified recipient 3. The system saves the published campaign
Alternative Flows	If the recipients or the sender emails are not real or do not exist, the system should notify the user.
Postconditions	The email has been sent to the specified recipient and the campaign has been saved.

Table 3.5: Use Case 4: Create Campaign

Use Case 5: Create Audience

Table 3.6 presents a textual description of a use case: Create Audience.

Actor	User, Admin
Goal	To Create and publish a campaign
Preconditions	The user is authenticated
Basic Flow	1. The user fills the necessary fields 2. The system validates the entered details and saves the Audience
Alternative Flows	If the user enters non valid audience details.
Postconditions	The Audience is created.

Table 3.6: Use Case 5: Create Audience

3.3 Design

This section provides an overview of the design aspects of our tool during the first sprint. It includes diagrams that illustrate the system's structure and behavior.

3.3.1 Class Diagram

The class diagram (Figure 3.2) visualizes the system's structure. It shows the classes, their attributes and operations, and the relationships among them.

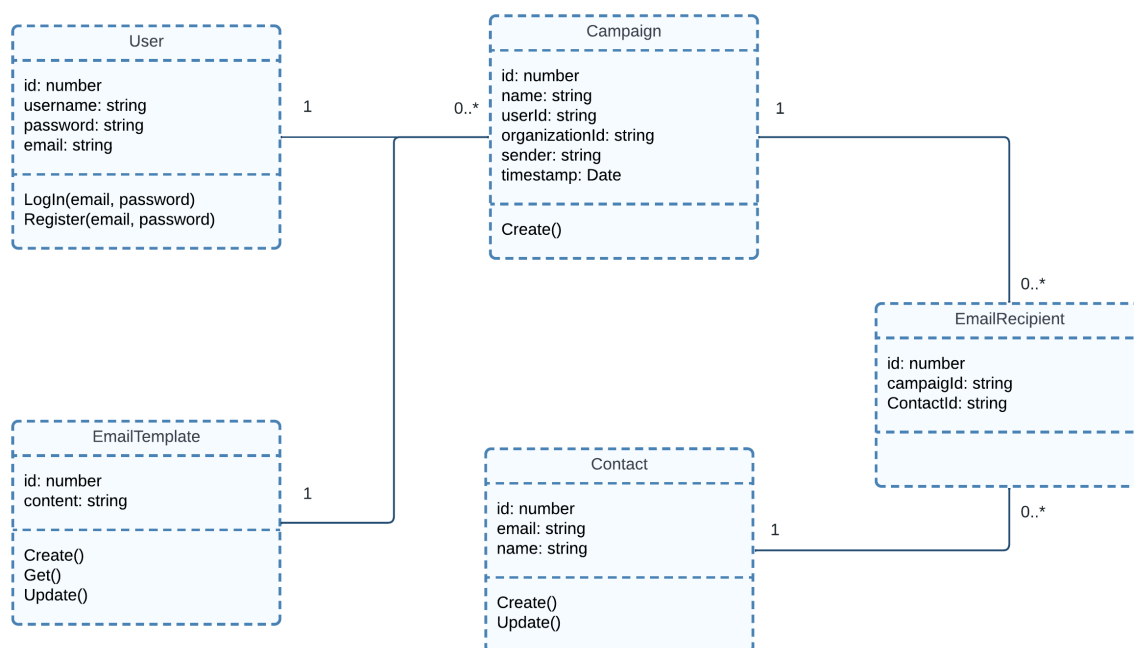


Figure 3.2: Sprint 1 Class Diagram

3.3.2 Sequence Diagrams

The sequence diagrams illustrate the flow of operations in the system, emphasizing the interaction between objects over time. Each diagram corresponds to a specific use case or functionality.

Log In Sequence Diagram

Figure 3.3 describes the scenario of the "Log In" use case.

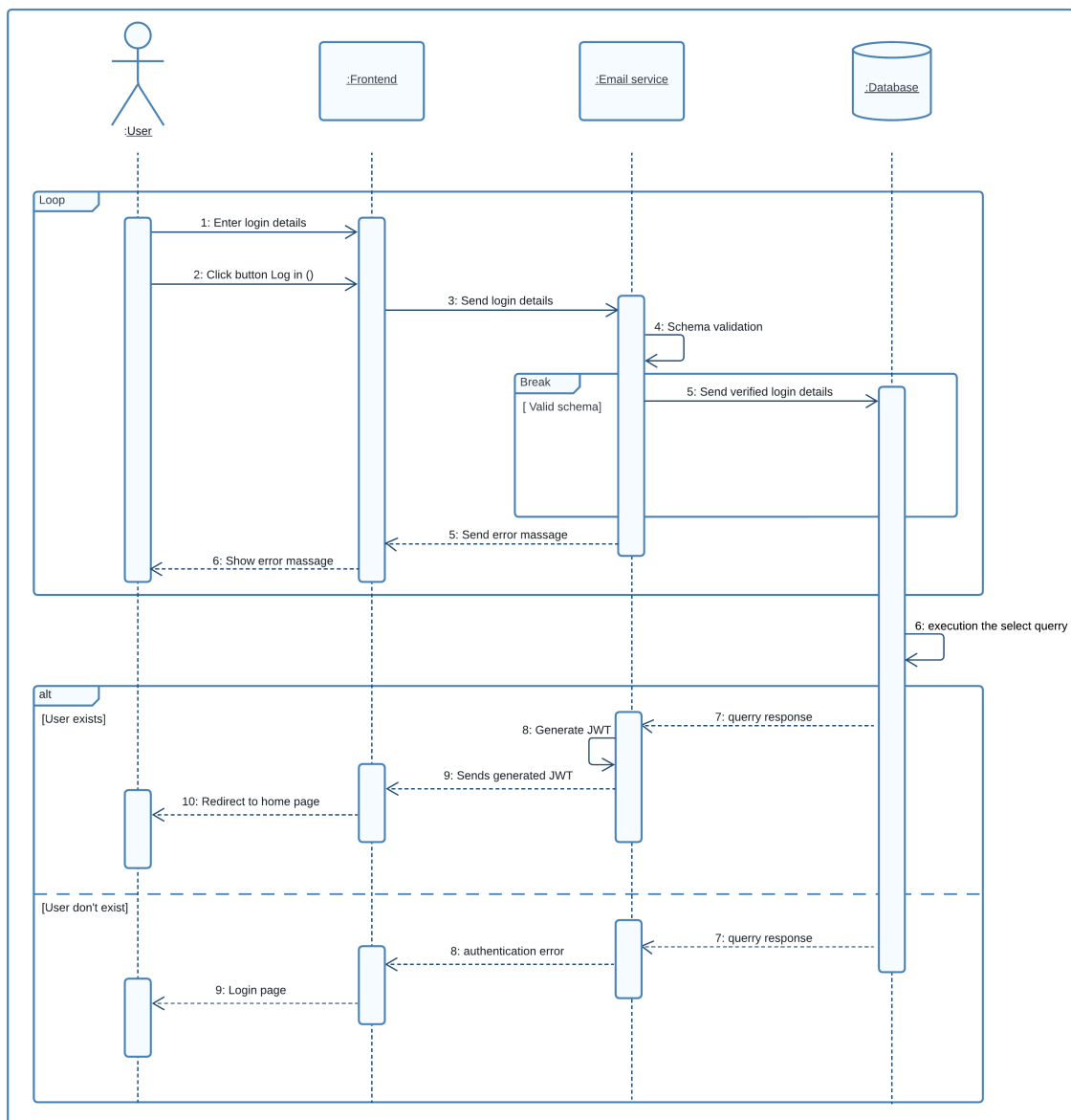


Figure 3.3: Log In Sequence Diagram

Register Sequence Diagram

Figure 3.4 describes the scenario of the "Register" use case.

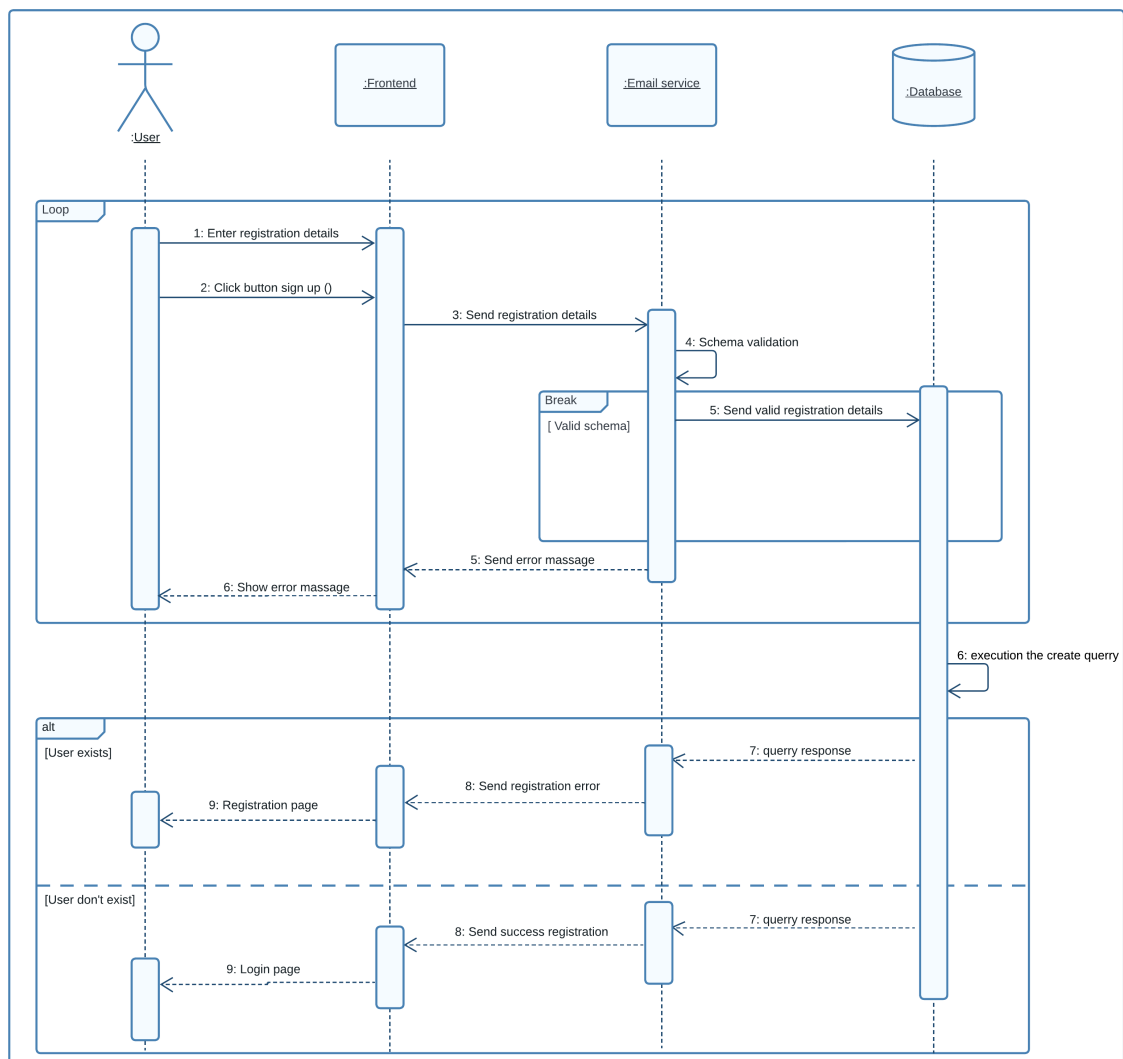


Figure 3.4: Register Sequence Diagram

Create Email Template Sequence Diagram

Figure 3.5 describes the scenario of the "Create Email Template" use case.

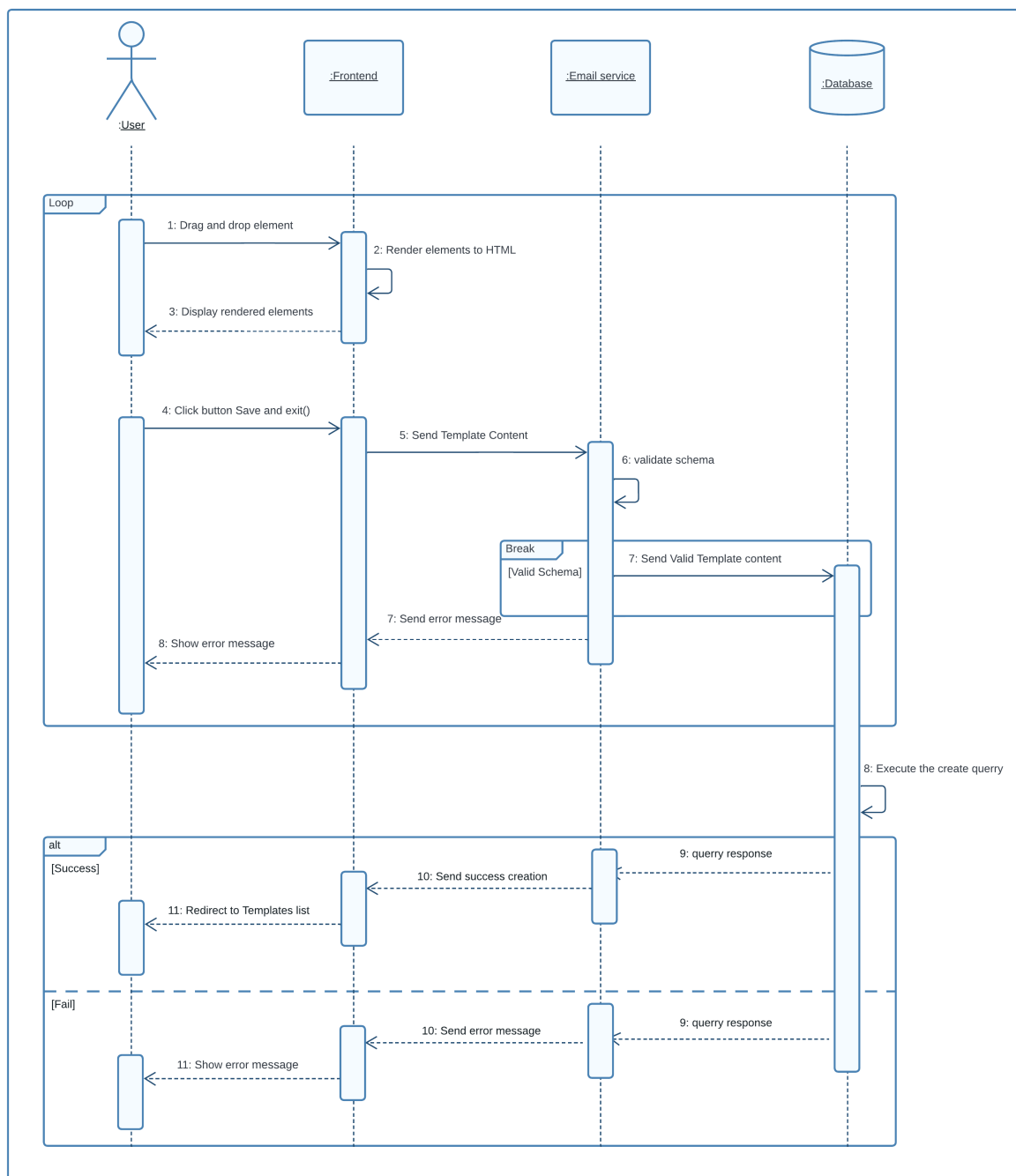


Figure 3.5: Create Email Template Sequence Diagram

Create And Publishing Campaigns Sequence Diagram

Figure 3.6 describes the scenario of the "Create And Publishing Campaigns" use case.

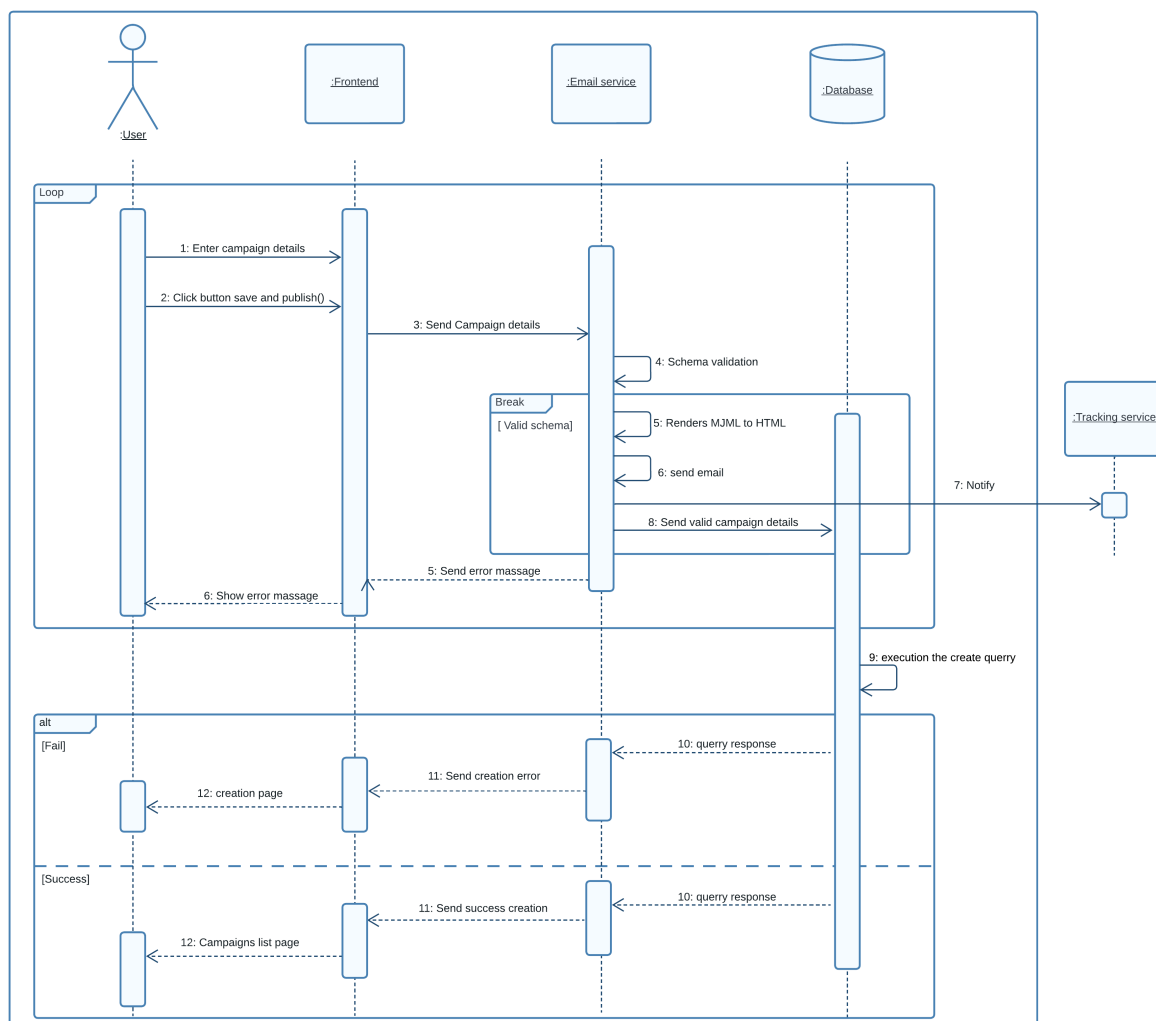


Figure 3.6: Create And Publishing Campaigns Sequence Diagram

Create Audience Sequence Diagram

Figure 3.7 describes the scenario of the "Create Audience" use case.

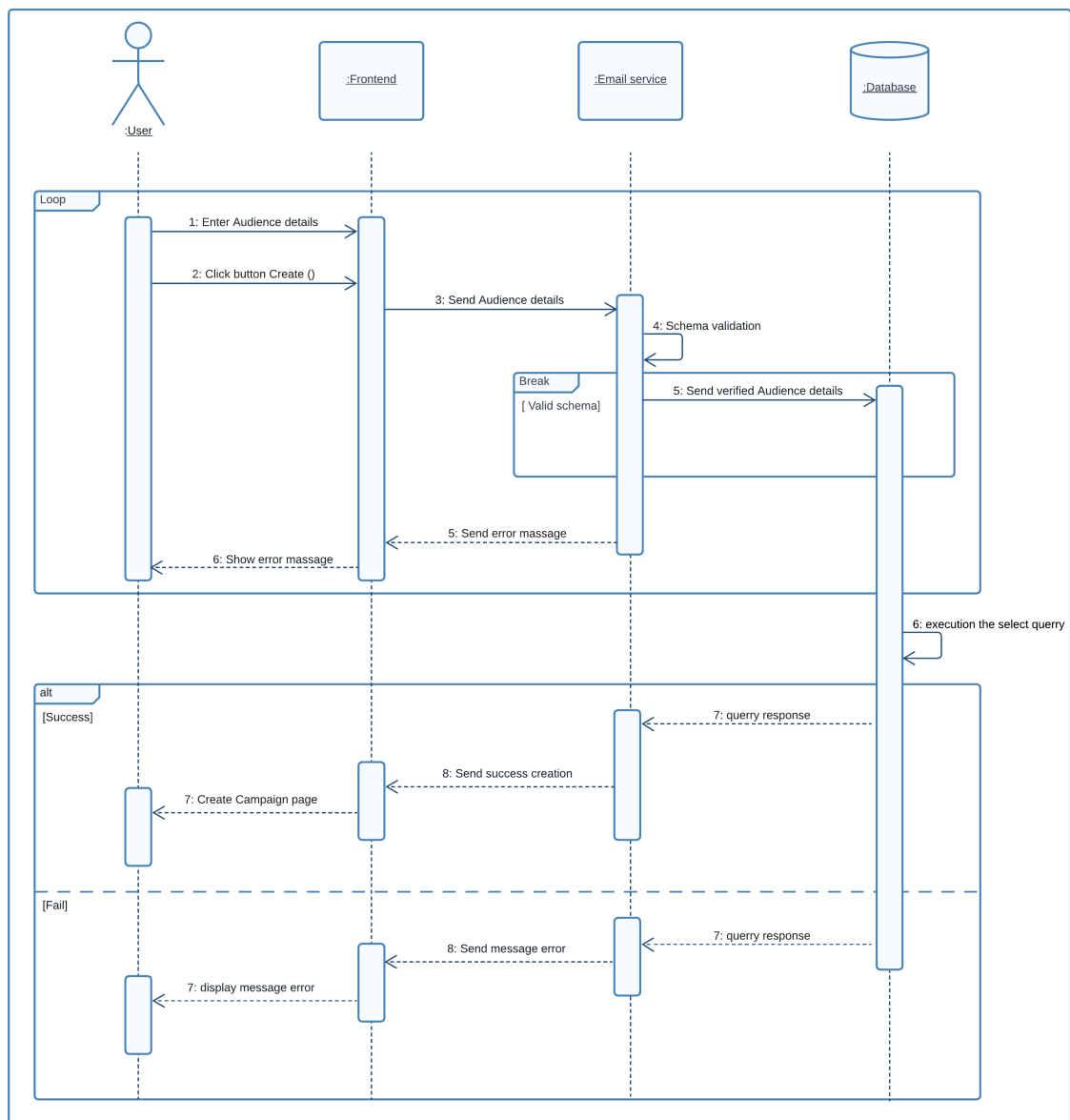


Figure 3.7: Create Audience Sequence Diagram

3.4 Implementation

This section provides a comprehensive overview of the system's implementation. It includes the user interface illustrating each component and feature implemented in our email marketing tool during the first sprint. The user interface was designed to be simple and user-friendly. The goal was to make the tool easy to use even for users without much technical expertise

3.4.1 Authentication Interface

For the authentication interface, I decided to use JSON Web Tokens (JWT) for authentication. This decision was made due to JWT's ability to securely transmit information between parties as a JSON object. This information can be verified and trusted because it is digitally signed.

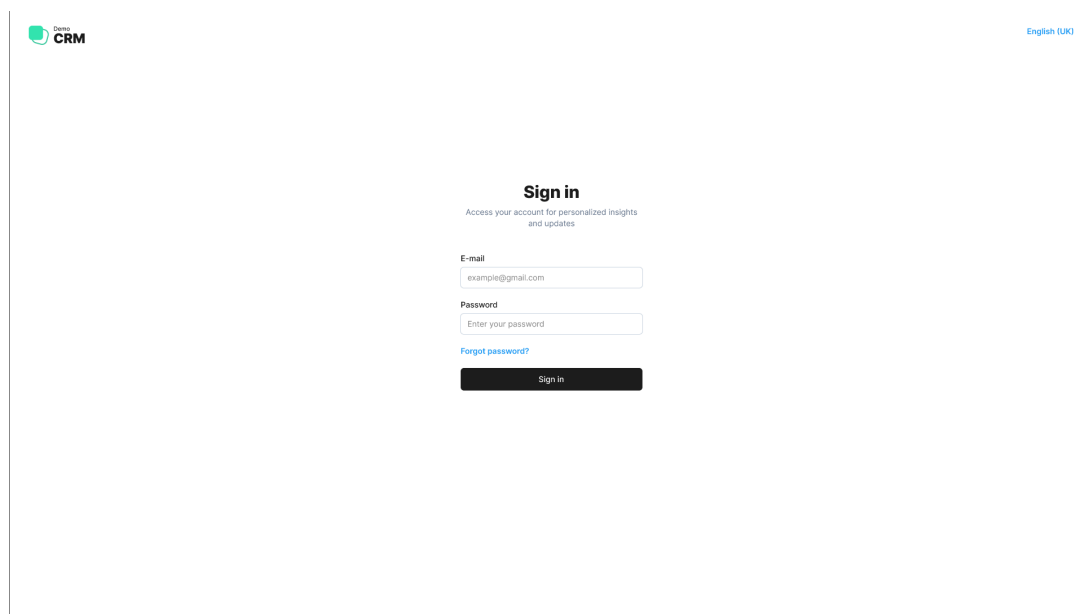


Figure 3.8: User Interface for Authentication Interface

3.4.2 Email Template Builder

The email template builder was one of the main features implemented in Sprint 1. I used a drag-and-drop interface for MJML blocks, which allows users to easily design their email templates. The MJML blocks are then rendered into HTML for compatibility with various email clients. This approach provides a balance between ease of use for the user

and flexibility in email design.

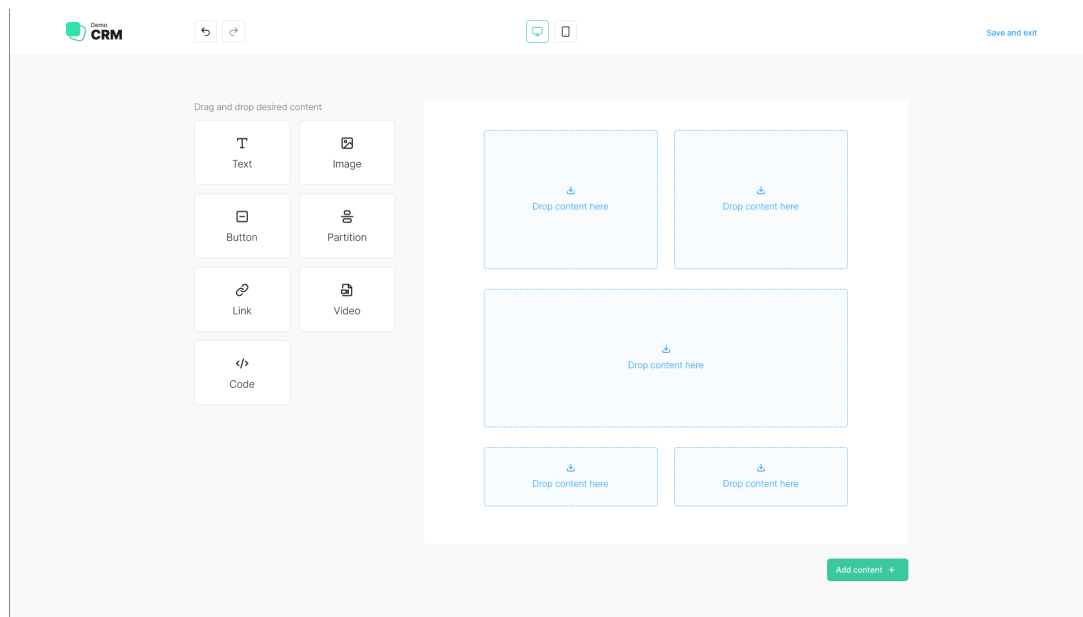
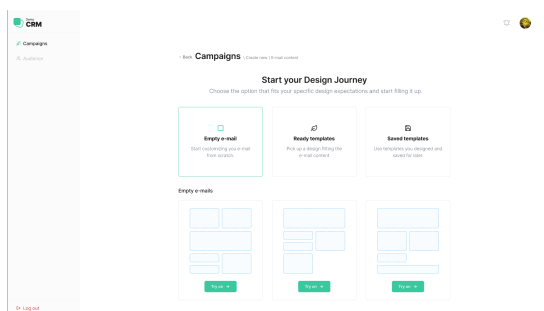


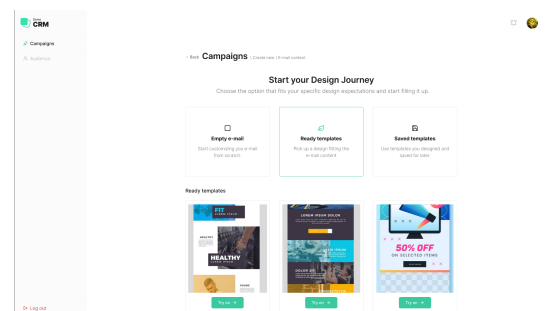
Figure 3.9: User Interface for Email Template Builder

3.4.3 Email Template Library

The email template library allows users to save and reuse email templates. Users can create new templates from scratch or save existing templates for future use. This feature is designed to save time and effort for users who frequently send similar emails.



(a) User Interface for Empty Email Templates Library



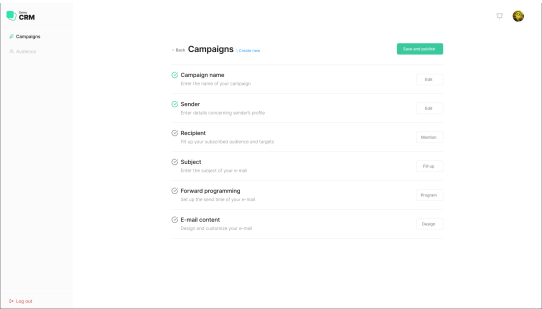
(b) User Interface for Ready Email Templates library

Figure 3.10: Email Template Library

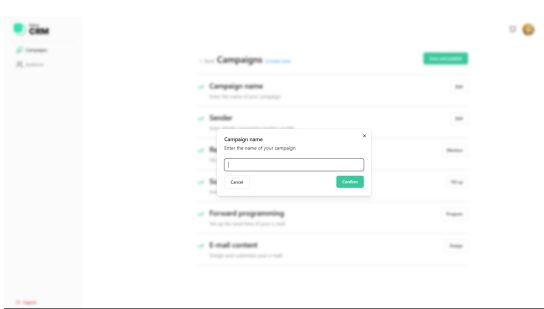
3.4.4 Campaigns Publishing and Creation

The functionality for creating and publishing campaigns was also implemented in this sprint. Users can create a campaign, design an email using the template builder, and

then publish the campaign to send the email to their audience.



(a) User Interface for Campaigns Publishing and Creation



(b) User Interface for Campaign creation form

Figure 3.11: User Interface for Campaigns Publishing and Creation

3.4.5 Campaigns List

The campaigns listing interface allows users to view and manage their existing campaigns. Users can see a list of all active and past campaigns, along with relevant details such as campaign name, send date, and updating date.

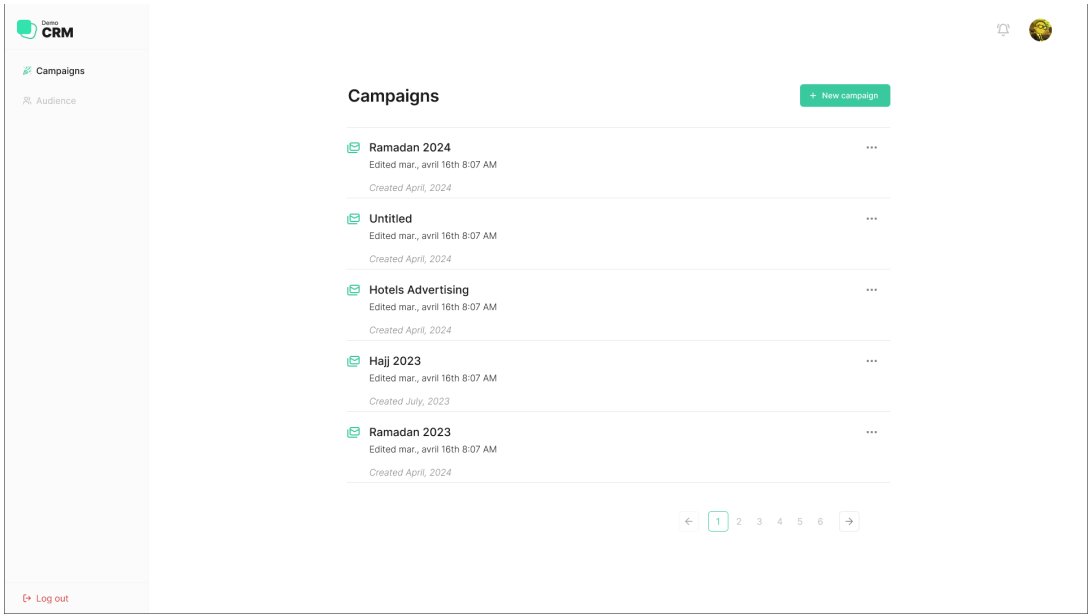


Figure 3.12: User Interface for Campaigns List

Conclusion

In this chapter, we outlined the Sprint 1 Backlog for our email marketing tool project. The tasks focus on developing fundamental features like user registration, authentication, email builder capabilities, and email sending functionalities. Prioritizing these components aims to establish a solid foundation for the application, enabling users to efficiently create, customize, and manage email campaigns. The structured approach and clear task delineation will facilitate effective teamwork and progress tracking, setting the stage for successful project execution and subsequent development phases.

4

Study and Implementation of Sprint 2

Introduction

This chapter provides a comprehensive look into the last sprint of our project. We begin by defining the tasks in our Backlog, followed by a detailed Functional Specification which includes a Use Case Diagram and a Textual Description of Use Cases. We then delve into the Design aspect, providing Detailed Sequence Diagrams and a Class Diagram. The Implementation section discusses the creation of an Event Dashboard Interface.

4.1 Sprint 2 Backlog

The table outlines the tasks for Sprint 2 related to organization creation, workspace setup, backend services, user interface, and email invitations.

ID	Task Description
1	Implement organization creation functionality, allowing users to create and manage different organizations. Each organization will have its own workspace.
2	Set up a workspace structure where shared data, such as published campaigns, media files, and email templates, can be accessed by authorized users within the organization.
3	Develop the backend services to handle organization-specific data and permissions.
4	Create an intuitive user interface for managing organization settings and workspace access.
5	Enable sending invitations via email to invite users to join an organization's workspace.

Table 4.1: Sprint 2 Backlog

The structured approach outlined in this backlog ensures clarity in understanding the responsibilities and goals for this sprint.

4.2 Functional Specification

This section provides a comprehensive overview of the system's functionality. It includes a use case diagram illustrating the system's interactions and a detailed textual description of each use case.

4.2.1 Use Case Diagram

Figure 4.1 offers us an overview by presenting a visual description of the functional behaviour of our tool during the second sprint. This diagram sums up the interactions among the actors and the diverse use cases within the system.

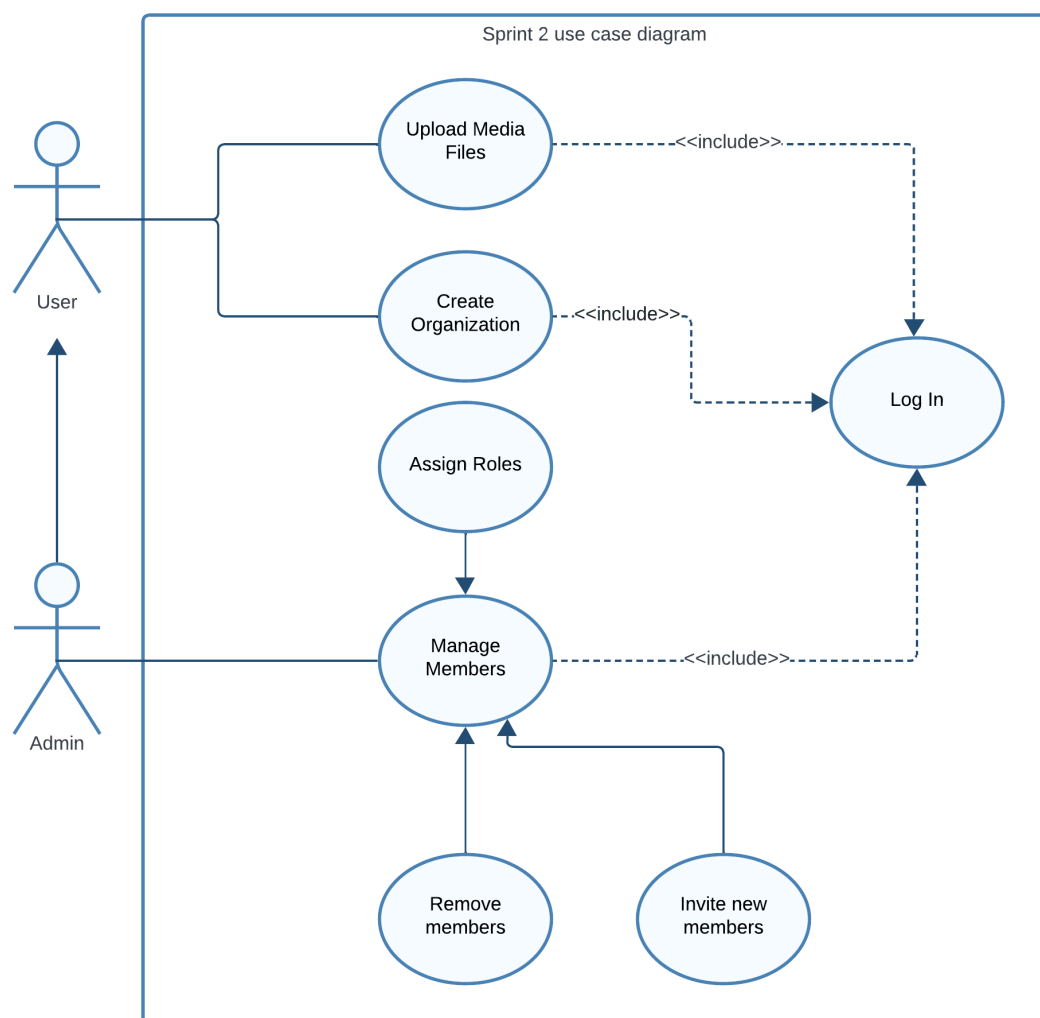


Figure 4.1: Sprint 2 Use Case Diagram

4.2.2 Textual Description of Use Cases

The following tables provide a detailed description of the use cases. This includes the actor, the goal of the use case, the preconditions, the basic flow, the alternative flows, and the postconditions.

Use Case 1: Upload Media Files

Table 4.2 presents a textual description of the use case: Upload Media Files.

Actor	User, admin
Goal	To upload media files (e.g., images, videos) to the system
Preconditions	The user is logged in and has access to the media upload feature
Basic Flow	1. The user selects the media files they want to upload 2. The system validates the file format and size 3. The system stores the uploaded files in the appropriate location
Alternative Flows	If the file format or size is invalid, the system displays an error message
Postconditions	The media files are successfully uploaded and associated with the user's account

Table 4.2: Use Case 1: Upload Media Files

Use Case 2: Create Organizations

Table 4.3 presents a textual description of the use case: Create Organizations.

Actor	User, Admin
Goal	To create new organizations within the system
Preconditions	The user or admin is logged in and has the necessary permissions to create organizations
Basic Flow	1. The user or admin selects the option to create a new organization 2. The system prompts for organization details 3. The user or admin provides the required information 4. The system validates the input and creates the organization
Alternative Flows	If the provided organization details are incomplete or invalid, the system shows an error message
Postconditions	A new organization is successfully created and added to the system

Table 4.3: Use Case 2: Create Organizations

Use Case 3: Assign Roles to Users

Table 4.4 presents a textual description of the use case: Assign Roles to Users.

Actor	Admin
Goal	To assign specific roles to users within an organization
Preconditions	The admin is logged in and has the necessary permissions to manage roles, targeted user already a member of the organization
Basic Flow	1. The admin choses a user within an organization 2. The system displays available roles (e.g., member, admin) 3. The admin assigns a role to the chosen user
Alternative Flows	If the admin selects an invalid user or the roles are not available, the system shows an error message
Postconditions	The user is assigned the specified roles within the organization

Table 4.4: Use Case 3: Assign Roles to Users

Use Case 4: Send Invitation to Join Organization

Table 4.5 presents a textual description of the use case: Send Invitation to Join Organization via Email.

Actor	Admin
Goal	To invite users to join an organization by sending email invitations
Preconditions	The admin is logged in and has the necessary permissions to manage organization invitations, the targeted user already have an account
Basic Flow	1. The admin enters users emails. 2. The system generates invitation emails with unique links. 3. The admin sends the invitation emails to the selected users.
Alternative Flows	If there are no eligible users to invite or if the email sending fails, the system shows an error message.
Postconditions	Invitation emails are successfully sent to the selected users.

Table 4.5: Use Case 4: Send Invitation to Join Organization via Email

Use Case 5: Remove User from an Organization

Table 4.6 presents a textual description of the use case: Remove User from an Organization.

Actor	Admin
Goal	To remove a user from an organization
Preconditions	The admin is logged in and has the necessary permissions to manage users within the organization. The targeted user is a member of the organization.
Basic Flow	1. The admin selects a user within the organization and clicks on the remove icon. 2. The system changes the status of the selected user to inactive and removes their association with the organization.
Alternative Flows	If the removal process fails due to a system error, the system notifies the admin and logs the error.
Postconditions	The user is successfully removed from the organization, and their status is updated to inactive.

Table 4.6: Use Case 5: Remove User from an Organization

4.3 Design

This section provides an overview of the design aspects of our tool during the second sprint. It includes diagrams that illustrate the system's structure and behavior.

4.3.1 Class Diagram

The class diagram (Figure 4.2) visualizes the system's structure. It shows the classes, their attributes and operations, and the relationships among them.

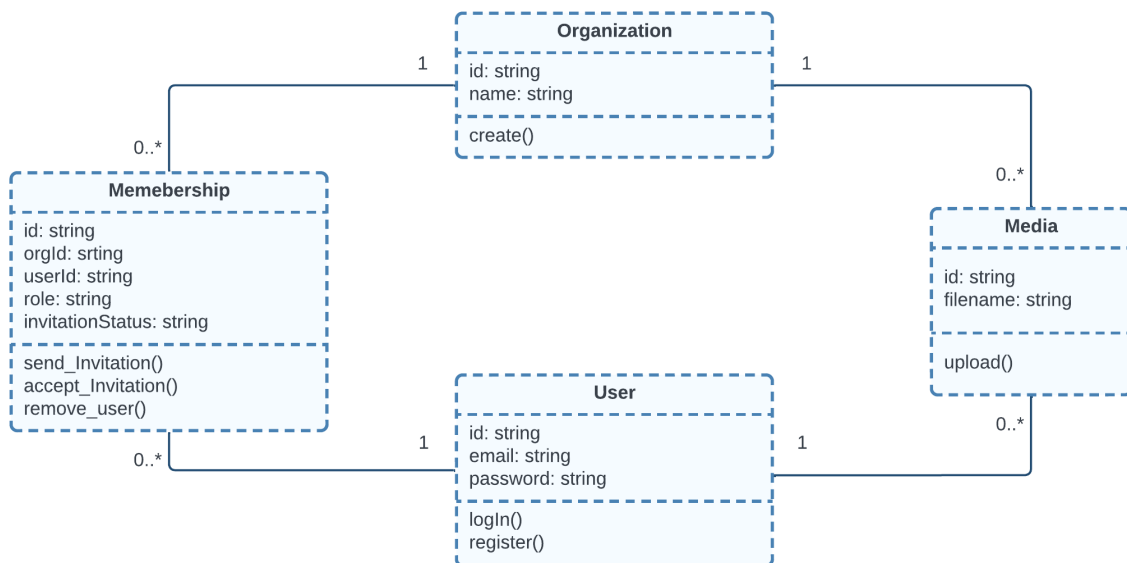


Figure 4.2: Sprint 2 Class Diagram

4.3.2 Sequence Diagrams

The sequence diagrams illustrate the flow of operations in the system, emphasizing the interaction between objects over time. Each diagram corresponds to a specific use case or functionality.

Media Upload Sequence Diagram

Figure 4.3 describes the scenario of the "Media Upload" use case.

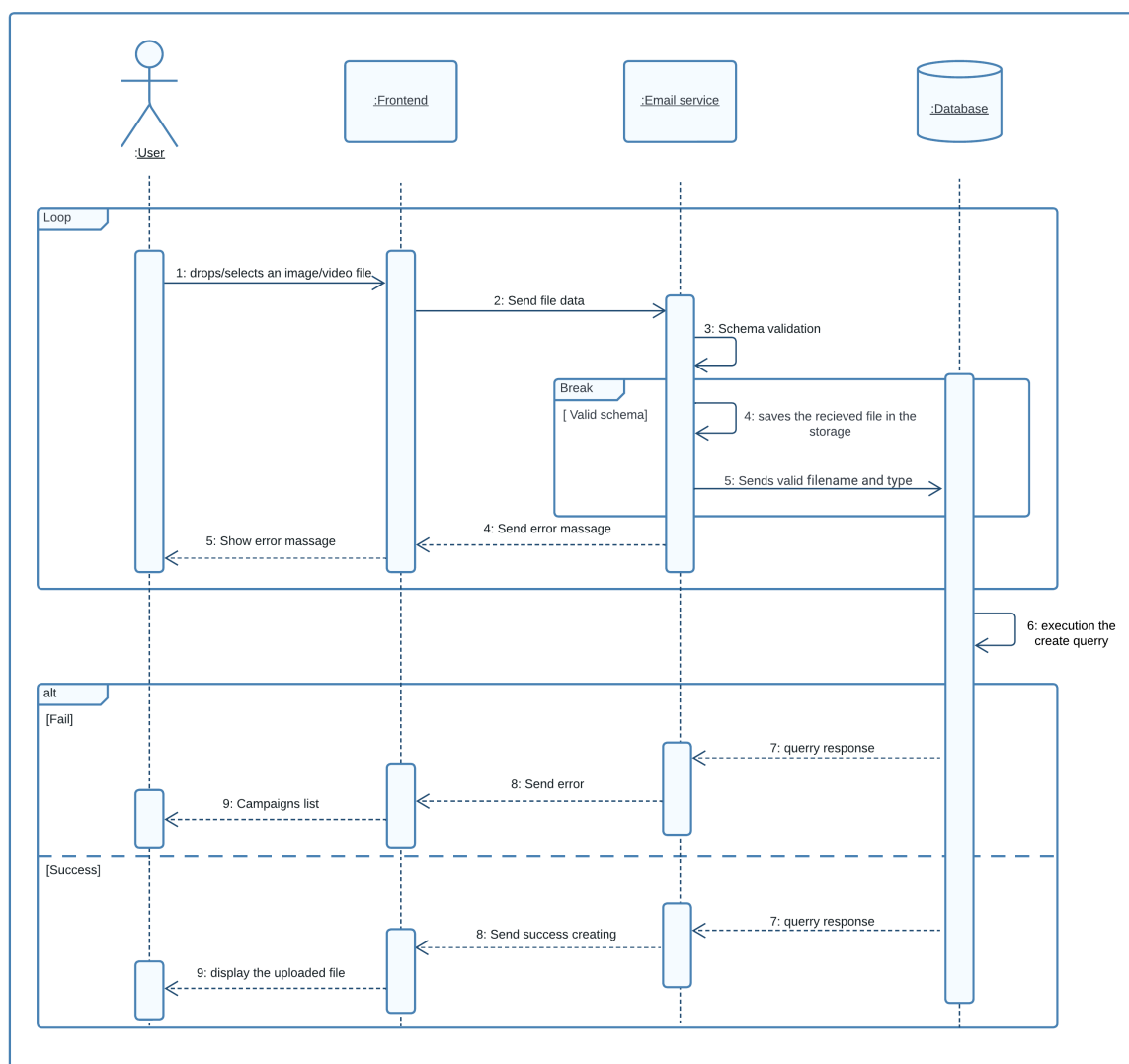


Figure 4.3: Media Upload Sequence Diagram

Organization Creation Sequence Diagram

Figure 4.4 describes the scenario of the "Organization Creation" use case.

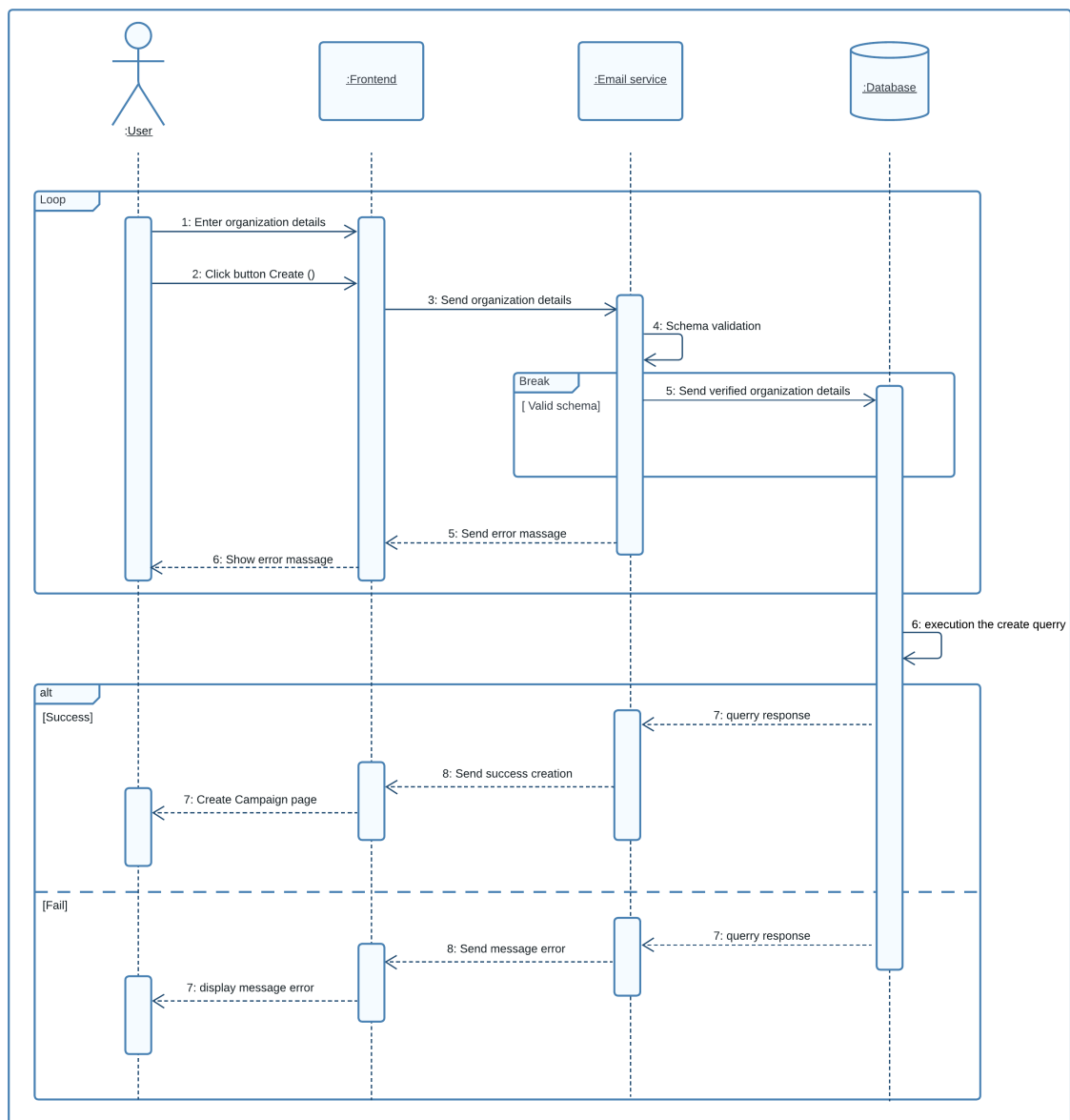


Figure 4.4: Organization Creation Sequence Diagram

Send Invitation Sequence Diagram

Figure 4.5 describes the scenario of the "Send Invitation" use case.

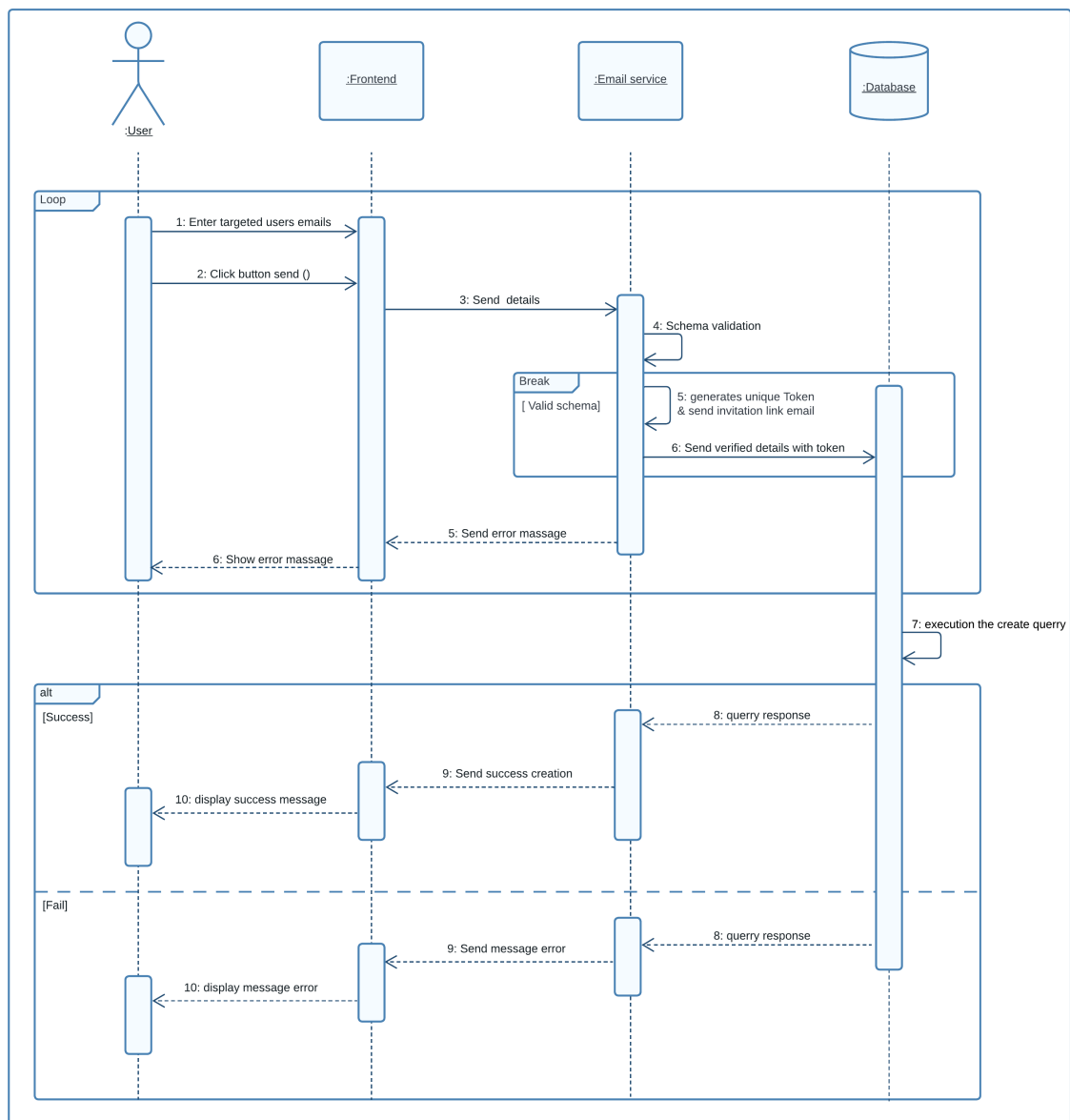


Figure 4.5: Send Invitation Sequence Diagram

Accept Invitation Sequence Diagram

Figure 4.6 describes the scenario of the "Accept Invitation" use case.

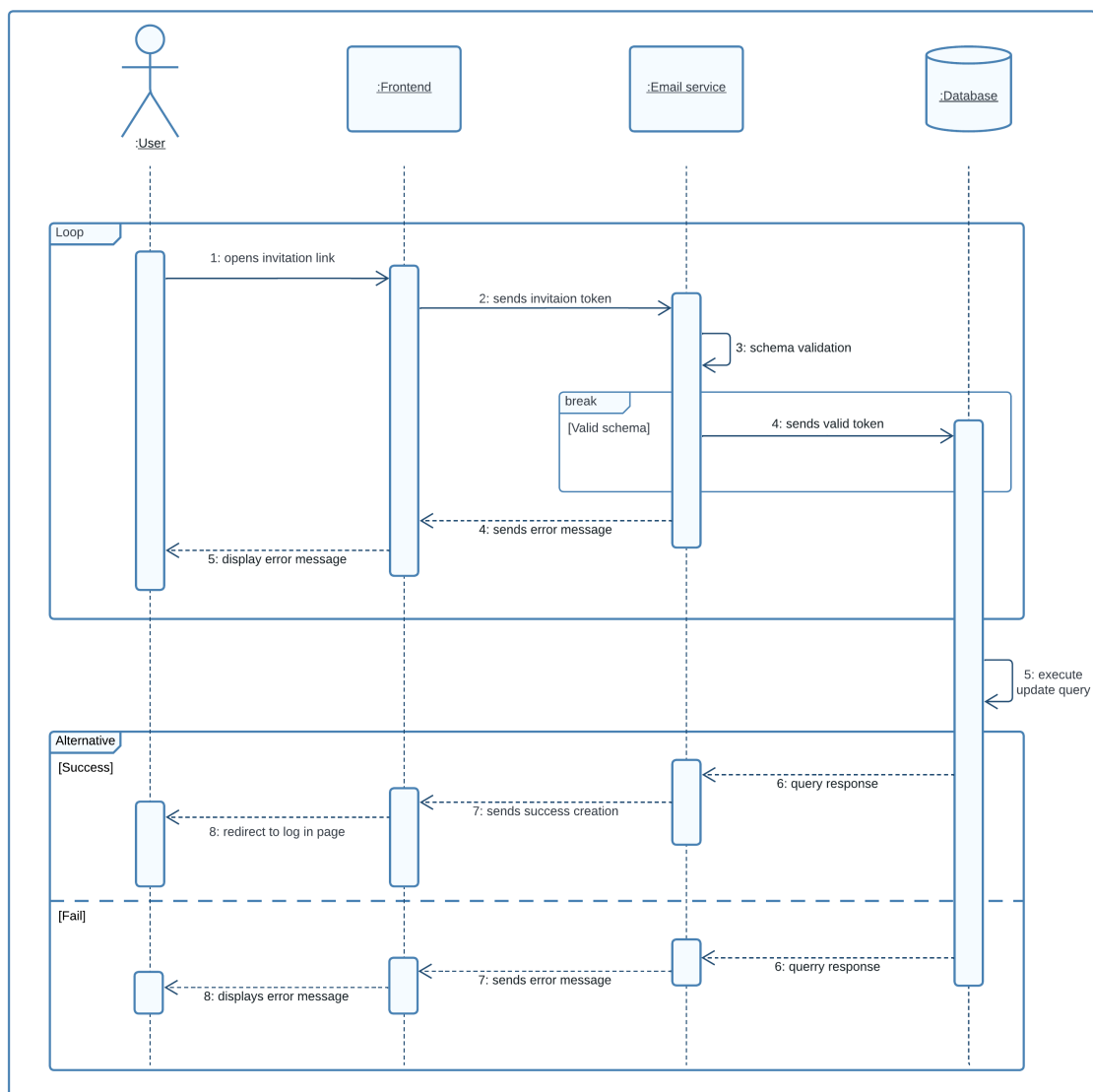


Figure 4.6: Accept Invitation Sequence Diagram

Assign Roles Sequence Diagram

Figure 4.7 describes the scenario of the "Assign Roles" use case.

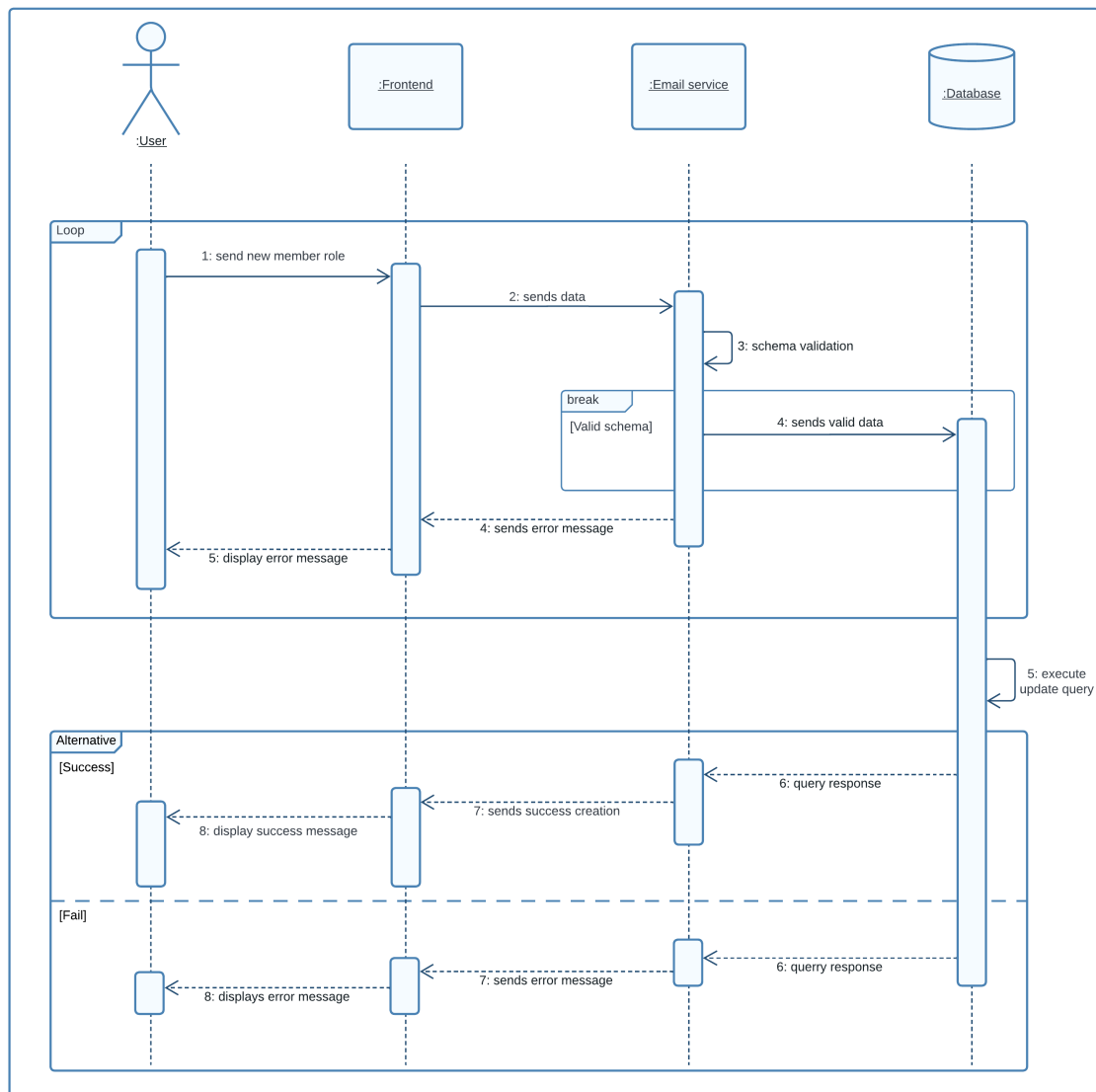


Figure 4.7: Assign Roles Sequence Diagram

Remove Members Sequence Diagram

Figure 4.8 describes the scenario of the "Remove Members" use case.

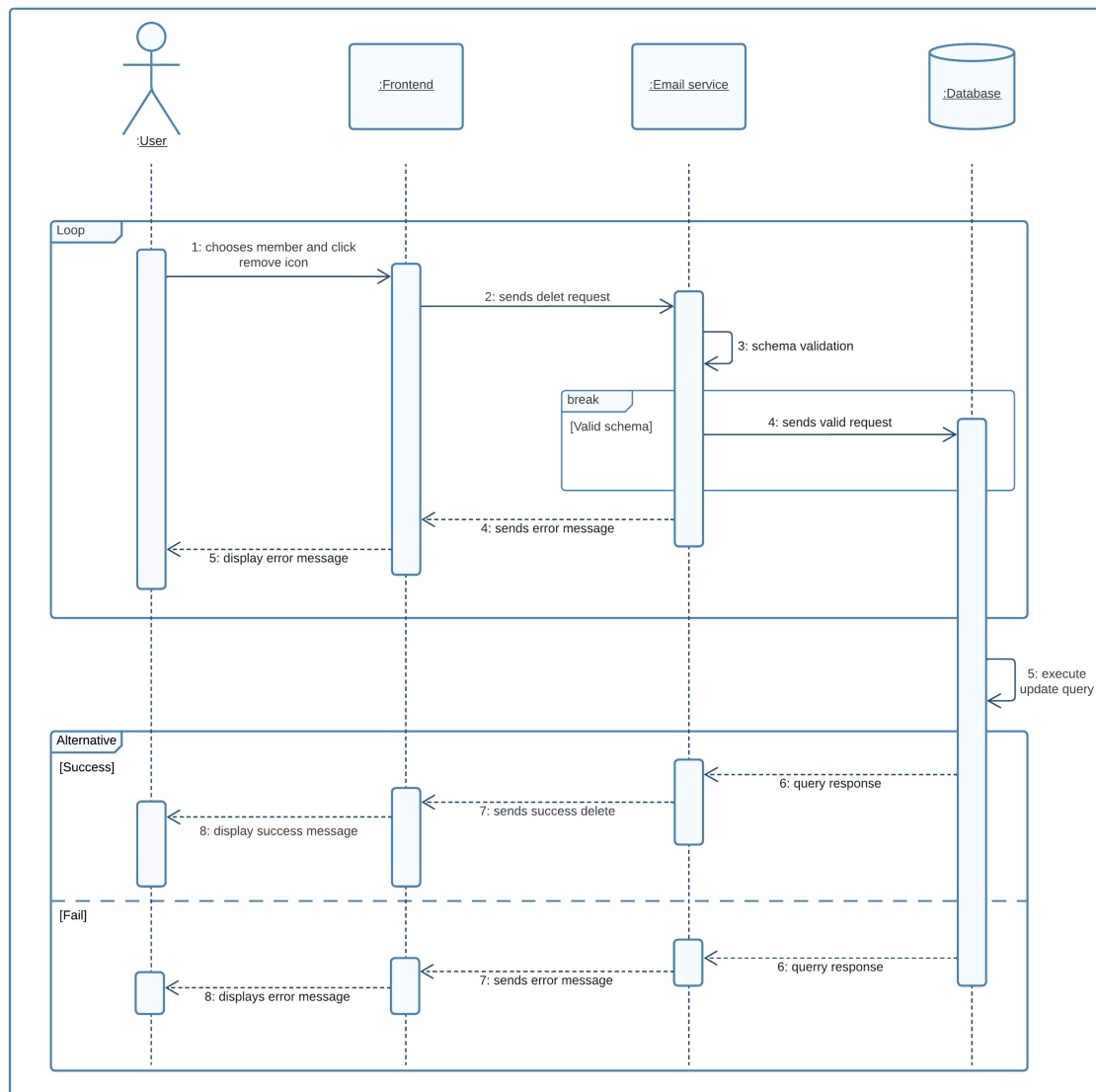


Figure 4.8: Remove Members Sequence Diagram

4.4 Implementation

This section provides a comprehensive overview of the system's implementation. It includes the user interface illustrating each component and feature implemented in our email marketing tool during the third sprint. The user interface was designed to be simple and user-friendly. The goal was to make the tool easy to use even for users without much technical expertise.

4.4.1 Upload Media

The *Upload Media* feature allows users to either drop a media file or select one to upload from their device. Once uploaded, these media files will be available in the email builder to create email templates with images. This feature ensures that users can easily access and use their media assets when designing their email campaigns.

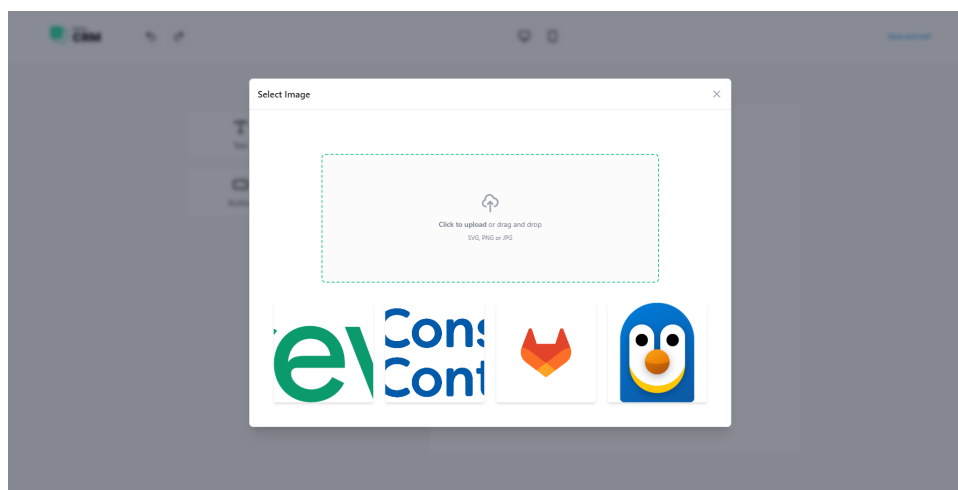


Figure 4.9: User Interface for Upload Media

4.4.2 Create New Organization

The *Create New Organization* feature functions as a workspace where every organization has its own campaigns, templates library, media cloud, and audience. By default, the creator of the organization becomes the admin. This setup ensures that each organization operates independently and securely with dedicated resources.



Figure 4.10: User Interface for Creating a New Organization

4.4.3 Invite New Members to Join Organization

The *Invite New Members* feature allows the admin of an organization to send an email invitation link with a unique token to targeted users to join the organization. Upon sending an invitation, the targeted user is saved in the database with a "PENDING" status. If the user clicks on the sent link, their status changes to "ACCEPTED". If the invitation is not accepted within a day, it will be declined.

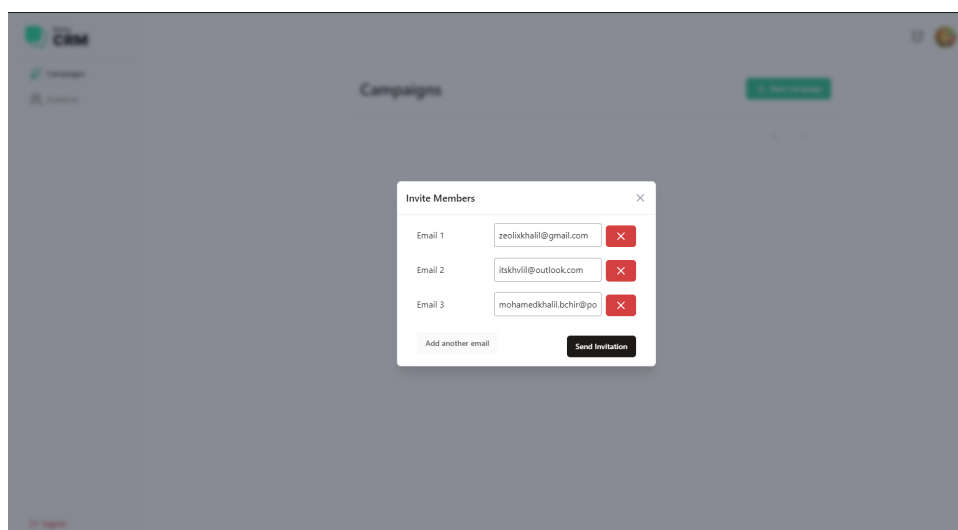


Figure 4.11: User Interface for Inviting New Members

4.4.4 Manage Members And Roles Within an Organization

Assign Roles The *Assign Roles* feature allows the admin to update the role of any member within the organization. Each role comes with specific permissions, enabling the admin to control the level of access and actions that different members can perform within the organization.

Remove Members The *Remove Members* feature enables the admin to remove a user from the organization by clicking on the remove icon. When a user is removed, their status is updated to "INACTIVE", ensuring that their data is not lost and can be reactivated if needed in the future.

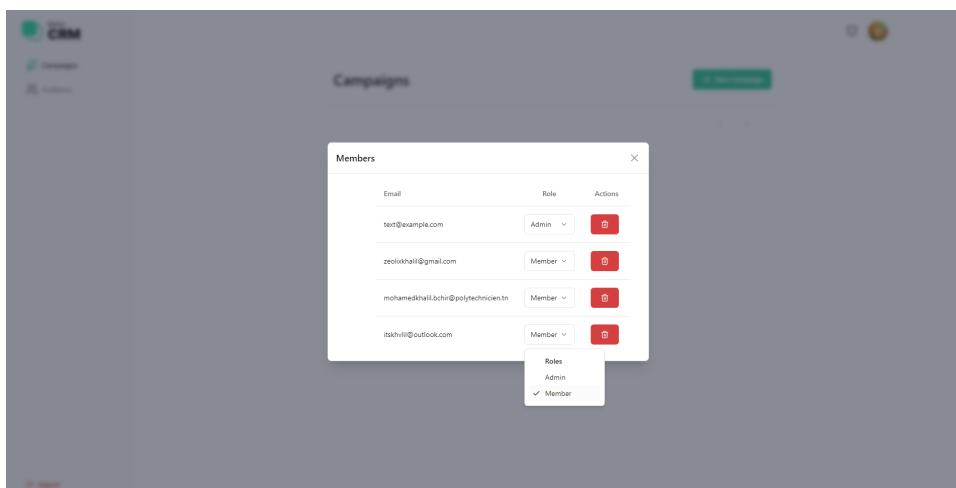


Figure 4.12: User Interface for Assigning Roles And Managing Members

Conclusion

In this chapter, we detailed the Sprint 2 Backlog for our email marketing tool project, focusing on enhancing organizational capabilities and collaboration. Key tasks include organization creation, workspace setup, backend services for data and permissions, a user-friendly interface for settings management, and email invitations for workspace access. These tasks aim to support effective team collaboration and resource sharing, enhancing overall functionality and user experience. This structured approach ensures clear team responsibilities and objectives, paving the way for successful project development and implementation.

5

Study and Implementation of Sprint 3

Introduction

This chapter provides a comprehensive look into the last sprint of our project. We begin by defining the tasks in our Backlog, followed by a detailed Functional Specification which includes a Use Case Diagram and a Textual Description of Use Cases. We then delve into the Design aspect, providing Detailed Sequence Diagrams and a Class Diagram. The Implementation section discusses the creation of an Event Dashboard Interface.

5.1 Sprint 3 Backlog

The table below outlines the tasks planned for Sprint 3. This sprint focuses on the development of key features for the tracking tool, specifically related to email and campaigns tracking, as well as CI/CD pipeline creation and deployment.

ID	Task Description
1	Create a CI/CD pipeline for automated deployment of new features.
2	Set up Docker for the backend services and database to run on the server.
3	Deploy the front-end application on Vercel for easy access and scalability.
4	Implement email tracking, allowing users to monitor email behavior (opens, clicks, etc.).
6	Create a dashboard that provides real-time insights into email performance and campaign effectiveness.

Table 5.1: Sprint 3 Backlog

The structured approach outlined in this backlog ensures clarity in understanding the responsibilities and goals for this sprint.

5.2 Functional Specification

This section provides a comprehensive overview of the system’s functionality. It includes a use case diagram illustrating the system’s interactions and a detailed textual description of each use case.

5.2.1 Use Case Diagram

Figure 5.1 offers us an overview by presenting a visual description of the functional behaviour of our tool during the third sprint. This diagram sums up the interactions among the actors and the diverse use cases within the system.

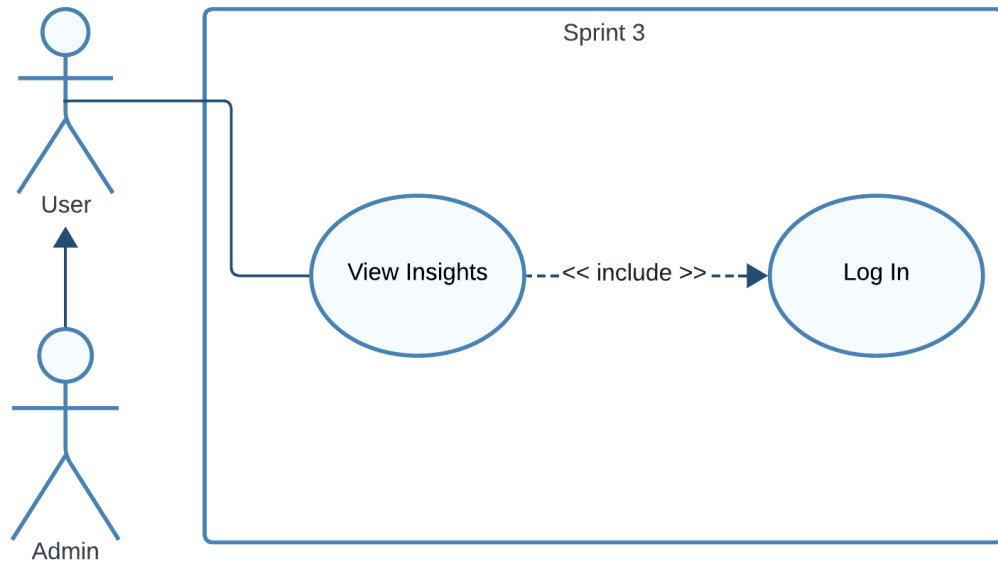


Figure 5.1: Sprint 3 Use Case Diagram

5.2.2 Textual Description of Use Cases

The following tables provide a detailed description of the use cases. This includes the actor, the goal of the use case, the preconditions, the basic flow, the alternative flows, and the postconditions.

Use Case 1 View Insights

Table 5.2 presents a textual description of a use case: View Insights.

Actor	User, Admin
Goal	To view insights of a published campaign
Preconditions	A campaign has been published and tracking data is available
Basic Flow	1. The user selects a published campaign from the list of campaigns. 2. The system retrieves and displays the insights details for the selected campaign.
Alternative Flows	If no tracking data is available, the system informs the user and prompts them to check back later.
Postconditions	The user has viewed the insights for the selected campaign.

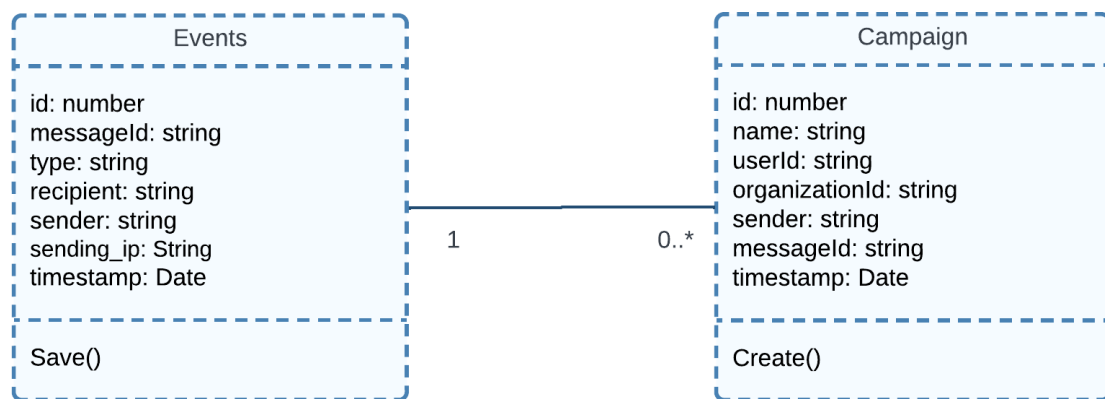
Table 5.2: Use Case 1: View Insights

5.3 Design

This section provides an overview of the design aspects of our tool during the third sprint. It includes diagrams that illustrate the system's structure and behavior.

5.3.1 Class Diagram

The class diagram (Figure 5.2) visualizes the system's structure. It shows the classes, their attributes and operations, and the relationships among them.

**Figure 5.2:** Sprint 3 Class Diagram

5.3.2 Sequence Diagrams

The sequence diagrams illustrate the flow of operations in the system, emphasizing the interaction between objects over time. Each diagram corresponds to a specific use case or functionality.

Saving Taracking Data Sequence Diagram Figure 5.3 describes the scenario of how the Tracking Data is being saved.

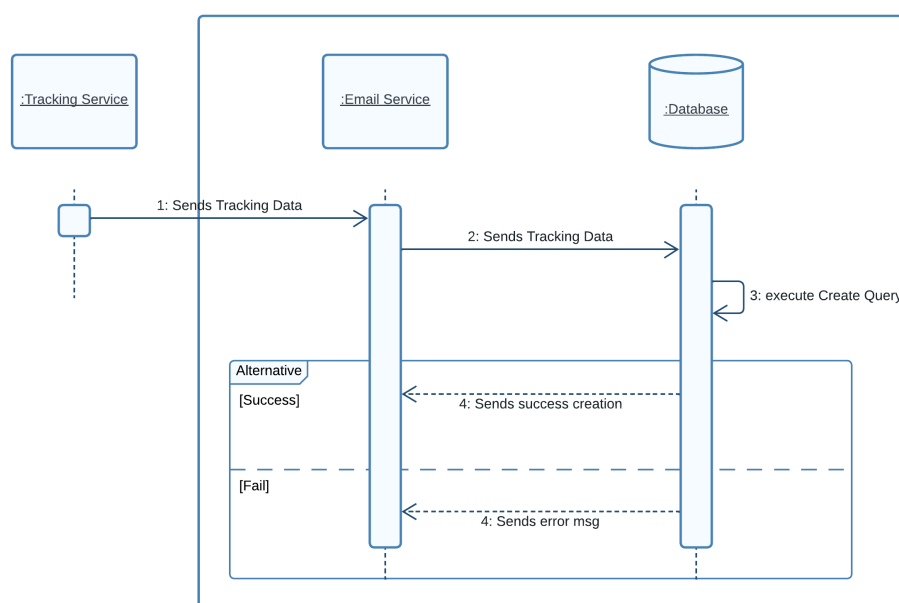


Figure 5.3: Tracking Sequence Diagram

Tracking Insights Sequence Diagram Figure 5.4 describes the scenario of how the Tracking Insights is being handled.

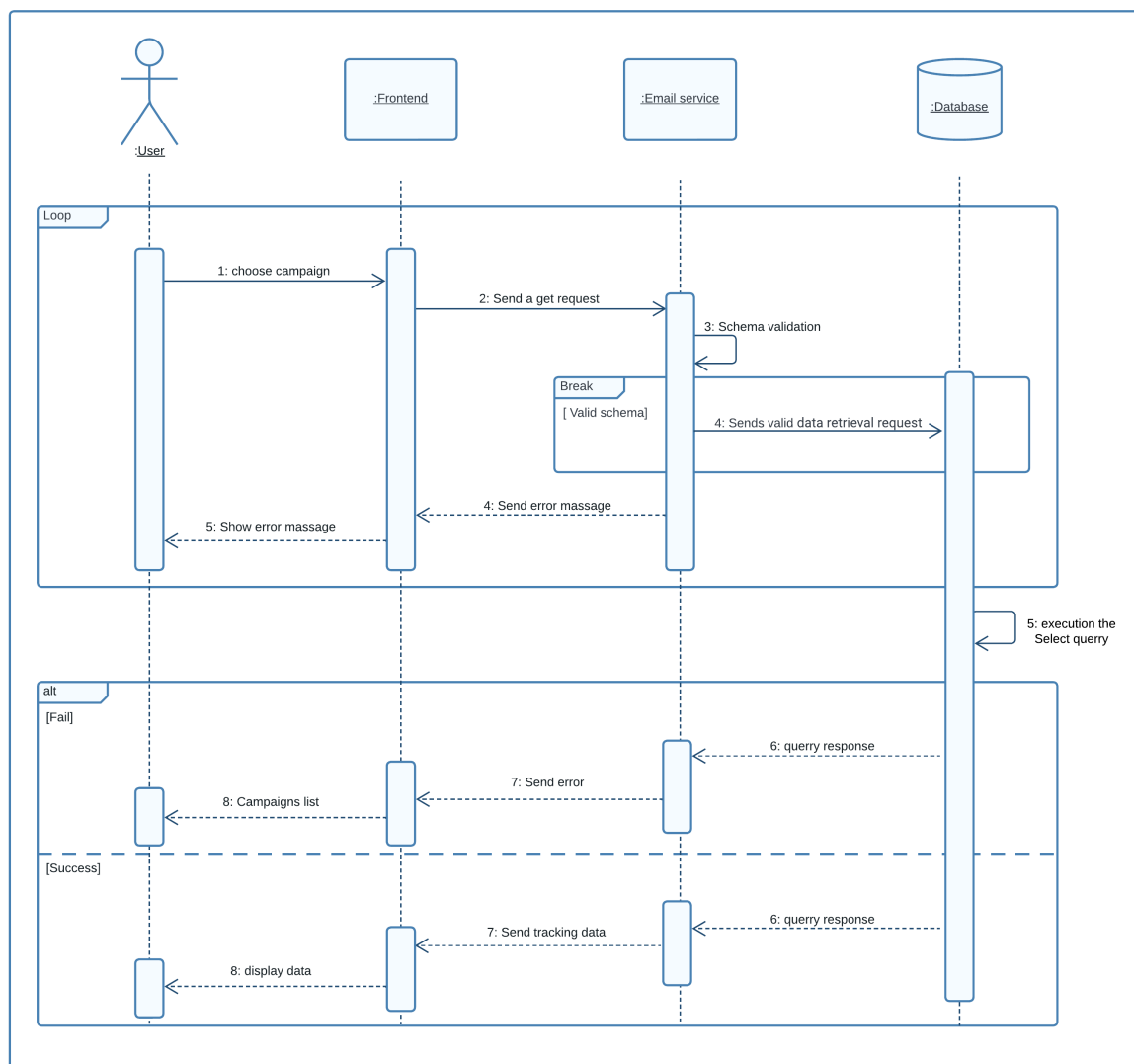


Figure 5.4: Tracking Sequence Diagram

5.4 Deployment

In this section, we'll cover the deployment process for your email marketing tool. We'll discuss setting up CI/CD pipelines, deploying backend services and the front-end deployment.

5.4.1 Deployment Process

1. CI/CD Pipeline Creation:

- Set up a CI/CD pipeline using GitLab CI/CD.
- Configure pipeline stages for building, deployment and database schema migration.

2. Backend Deployment:

- Containerize backend services using Docker.
- Create Dockerfiles for the email service.
- Define a 'docker-compose.yml' file for managing the email service and database containers.
- Deploy backend containers to a Azure.

3. Front-End Deployment:

- Deploy the Vue.js front-end application separately.
- Use Vercel for front-end deployment.
- Configure Vercel project settings environment variables.

5.4.2 Deployment Diagram

Figure 5.5 illustrates the deployment of our app.

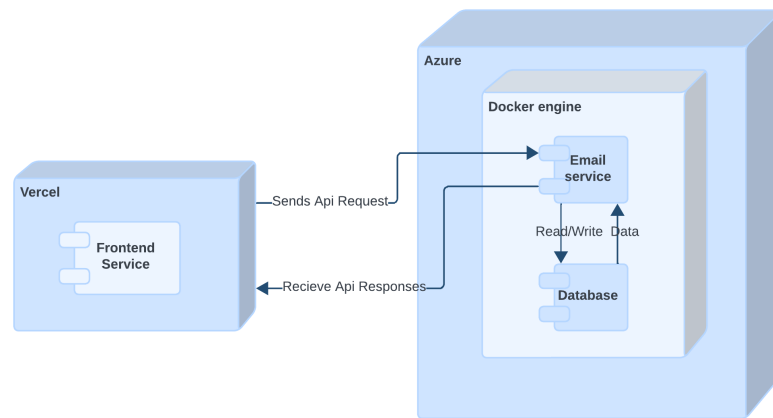


Figure 5.5: Deployment Diagram

5.4.3 CI/CD Pipeline Diagram

Figure 5.6 illustrates the stages of the CI/CD pipeline.

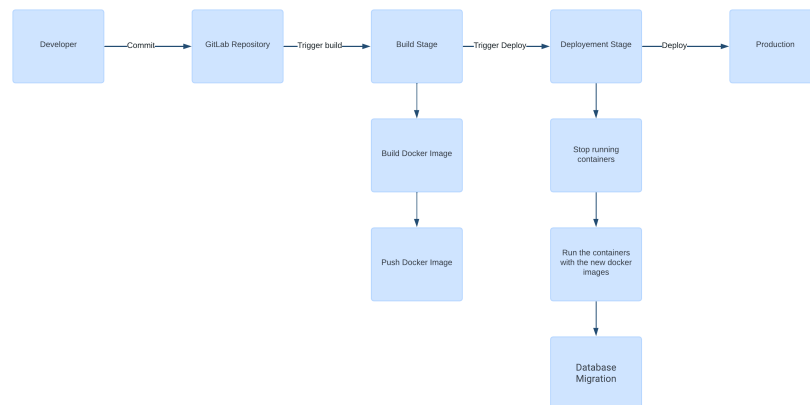


Figure 5.6: CI/CD Pipeline Diagram

5.5 Implementation

This section provides a comprehensive overview of the system's implementation. It includes the user interface illustrating each component and feature implemented in our

email marketing tool during the third sprint. The user interface was designed to be simple and user-friendly. The goal was to make the tool easy to use even for users without much technical expertise.

5.5.1 Insights Dashboard

Tracking Mechanism: To gather data for the Insights Dashboard, we rely on a tracking system. When an email campaign is sent, each email contains unique identifiers (Message ID). As recipients interact with the emails (e.g., open an email, click a link), these identifiers trigger events. Our system captures these events and records them in our database.

Brevo Transactional Webhook: We’ve integrated Brevo’s transactional webhook into our system. When specific events occur (e.g., email opened, link clicked), Brevo sends HTTP requests to our server. Our Email Service processes these requests, extracts relevant data (such as recipient email, campaign ID, event type), and updates our database. Real-time updates ensure that the Insights Dashboard reflects the latest campaign performance.

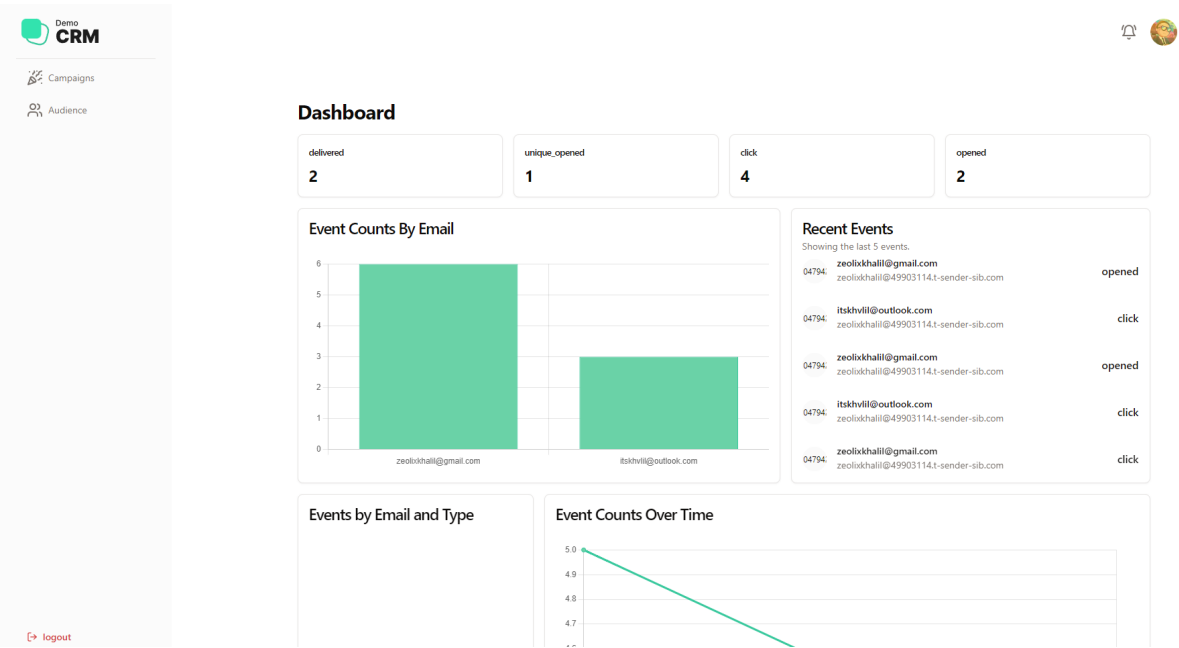


Figure 5.7: User Interface for Campaigns Dashboard

Conclusion

In this chapter, we outlined the Sprint 3 Backlog for our email marketing tool project. This sprint focuses on crucial tracking features and setting up the CI/CD pipeline for automated deployments. Key tasks include creating the CI/CD pipeline, setting up Docker for backend services and the database, deploying the front-end on Vercel, implementing email tracking, and developing a real-time performance dashboard. These tasks aim to enhance the application's reliability, scalability, and user experience. This structured approach ensures clarity in team roles and sprint objectives, facilitating efficient progress and successful milestone completion.

General Conclusion

Throughout this document, we meticulously outlined the backlogs for the initial three sprints of our email marketing tool project, accompanied by detailed diagrams, descriptions, and screenshots. Each sprint focused on developing essential features to build a robust application.

Sprint 1 established foundational features like user registration, authentication, and the email builder. The use case diagrams, textual descriptions, and implementation screenshots provided clarity on user interactions and system structure.

Sprint 2 enhanced organizational capabilities and collaboration through organization creation, workspace setup, and user invitations. Detailed diagrams and descriptions visualized the collaborative environment, while implementation screenshots showcased user interface improvements.

Sprint 3 implemented advanced tracking features and a CI/CD pipeline for automated deployment. Use case diagrams and descriptions clarified tracking processes, while design diagrams detailed technical aspects. Implementation screenshots highlighted tracking features and dashboard functionalities.

The structured approach in each sprint backlog, coupled with comprehensive documentation, ensures clarity in task responsibilities. This methodology fosters efficient teamwork and progress. We are poised to deliver a comprehensive email marketing tool meeting user needs and expectations.

References

- [1] Scrum guides. <https://scrumguides.org/scrum-guide.html>.
- [2] Visual studio code. <https://code.visualstudio.com/>.
- [3] Gitlab. <https://about.gitlab.com/>.
- [4] Notion. <https://www.notion.so/>.
- [5] Postman. <https://www.postman.com/>.
- [6] Dbeaver. <https://dbeaver.io/>.
- [7] Docker desktop. <https://www.docker.com/products/docker-desktop>.
- [8] Git. <https://git-scm.com/>.
- [9] Brevo. <https://developers.brevo.com/docs/transactional-webhooks>.
- [10] Node.js. <https://nodejs.org/>.
- [11] Typescript. <https://www.typescriptlang.org/>.
- [12] Vue 3. <https://vuejs.org/>.
- [13] Tailwind css. <https://tailwindcss.com/>.
- [14] Fastify. <https://www.fastify.io/>.
- [15] Mjml. <https://mjml.io/>.
- [16] Turborepo. <https://turbo.build/repo>.
- [17] Prisma. <https://www.prisma.io/>.
- [18] Swagger. <https://swagger.io/>.

Résumé:

Le présent travail s'inscrit dans le cadre du projet de fin d'études réalisé au sein de l'entreprise JoodLab en vue de l'obtention du Diplôme National d'Ingénieur en Génie Informatique à l'école Polytechnique Sousse pour l'année universitaire 2023/2024. Ce projet consiste à la Conception et développement d'un outil de e-mail marketing dynamique

Mots clés --- email, fastify, vue.js, sql

Summary:

This work is part of the end-of-studies project carried out within the company JoodLab, aiming to obtain the National Diploma in Computer Engineering at the Polytechnic School of Sousse for the academic year 2023/2024. This project involves the design and development of a dynamic email marketing tool.

Keywords --- email, fastify, vue.js, SQL



Campus Polytechnique Sousse,
Sahloul Sousse



(+216) 99 277 877



polytecsousse.tn